Bayesian Networks

Naïve Bayes Classifier

- What if we have multiple tests to decide is a person has cancer?
  - $P(A_i|B_1, B_2, \dots, B_n)$?

- Simplified assumption: attributes are conditionally independent

Naïve Bayes advantages

- Independence allows parameters to be estimated on different data sets, e.g.
  - Estimate content features from messages with headers omitted
  - Estimate header features from messages with content missing

Naïve Bayes disadvantages

- What if we have many more variables?
  - Want to execute different types of queries
  - Cannot assume independence between all variables

The Joint Probability Distribution

- Joint probabilities can be between any number of variables

  eg. *P(A = true, B = true, C = true)*
- For each combination of variables, we need to say how probable that combination is
- The probabilities of these combinations need to sum to 1

A	B	C	P(A,B,C)
false	false	false	0.1
false	false	true	0.2
false	true	false	0.05
false	true	true	0.05
true	false	false	0.3
true	false	true	0.1
true	true	false	0.05
true	true	true	0.15

Sums to 1

The Joint Probability Distribution

- Once you have the joint probability distribution, you can calculate **any** probability involving *A*, *B*, and *C*
- Only need
  - Marginalization, and
  - definition of conditional probability

A	B	C	P(A,B,C)
false	false	false	0.1
false	false	true	0.2
false	true	false	0.05
false	true	true	0.05
true	false	false	0.3
true	false	true	0.1
true	true	false	0.05
true	true	true	0.15

Examples of things you can compute:

- $P(A\text{=true})$ = sum of $P(A,B,C)$ in rows with A=true
- $P(A\text{=true}, B = \text{true} \mid C\text{=true}) = \dfrac{P(A\text{=true}, B = \text{true}, C\text{=true})}{P(C\text{=true})}$

Marginalization

- To get probability ***without*** a variable *A*:
  - Sum up all rows containing different values of *A*, without changing other variable values

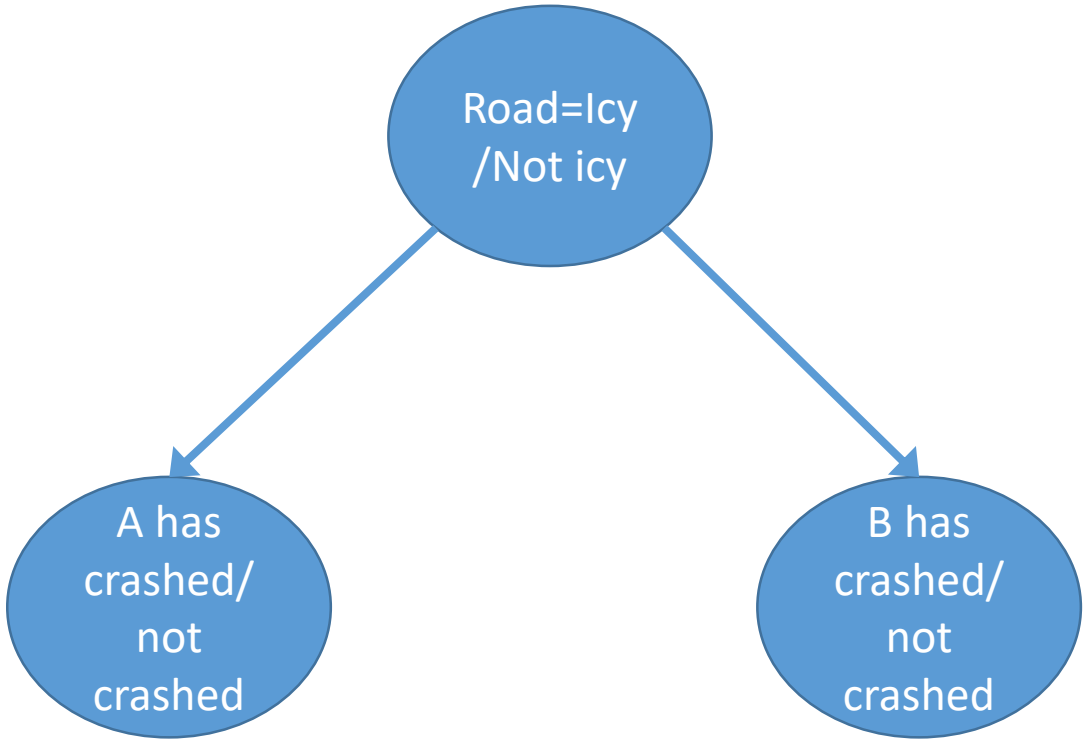
The Problem with the Joint Distribution

- Lots of entries in the table
- For *k* Boolean random variables, table of size 2^k
- How do we use fewer numbers?
- Exploit the concept of ***conditional independence***

A	B	C	P(A,B,C)
false	false	false	0.1
false	false	true	0.2
false	true	false	0.05
false	true	true	0.05
true	false	false	0.3
true	false	true	0.1
true	true	false	0.05
true	true	true	0.15

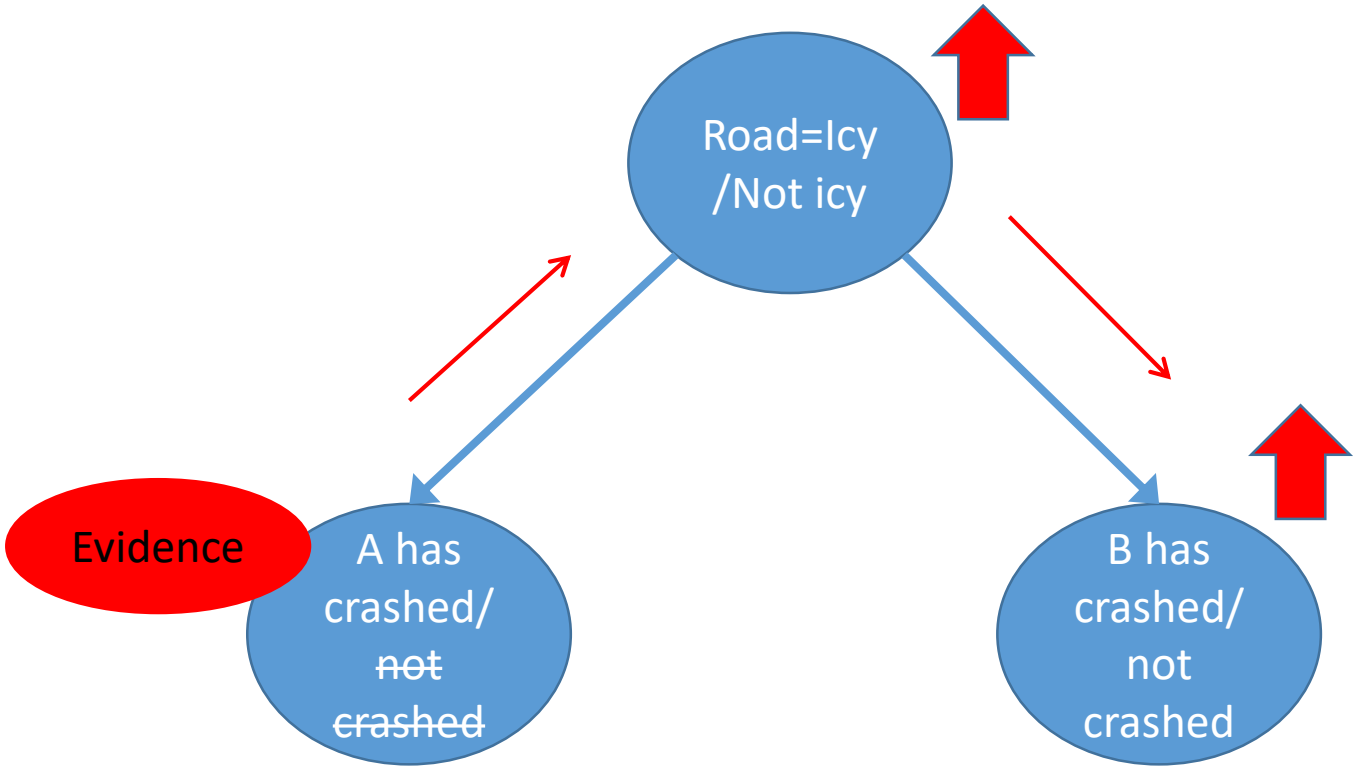
“Icy roads” example

- A road can be icy or not
- There is a chance that driving on that road will result in an accident
  - Accident is much more likely if road is icy
- Consider two people, A and B, driving on that road


Road=Icy
/Not icy
A has
crashed/
not
crashed
B has
crashed/
not
crashed

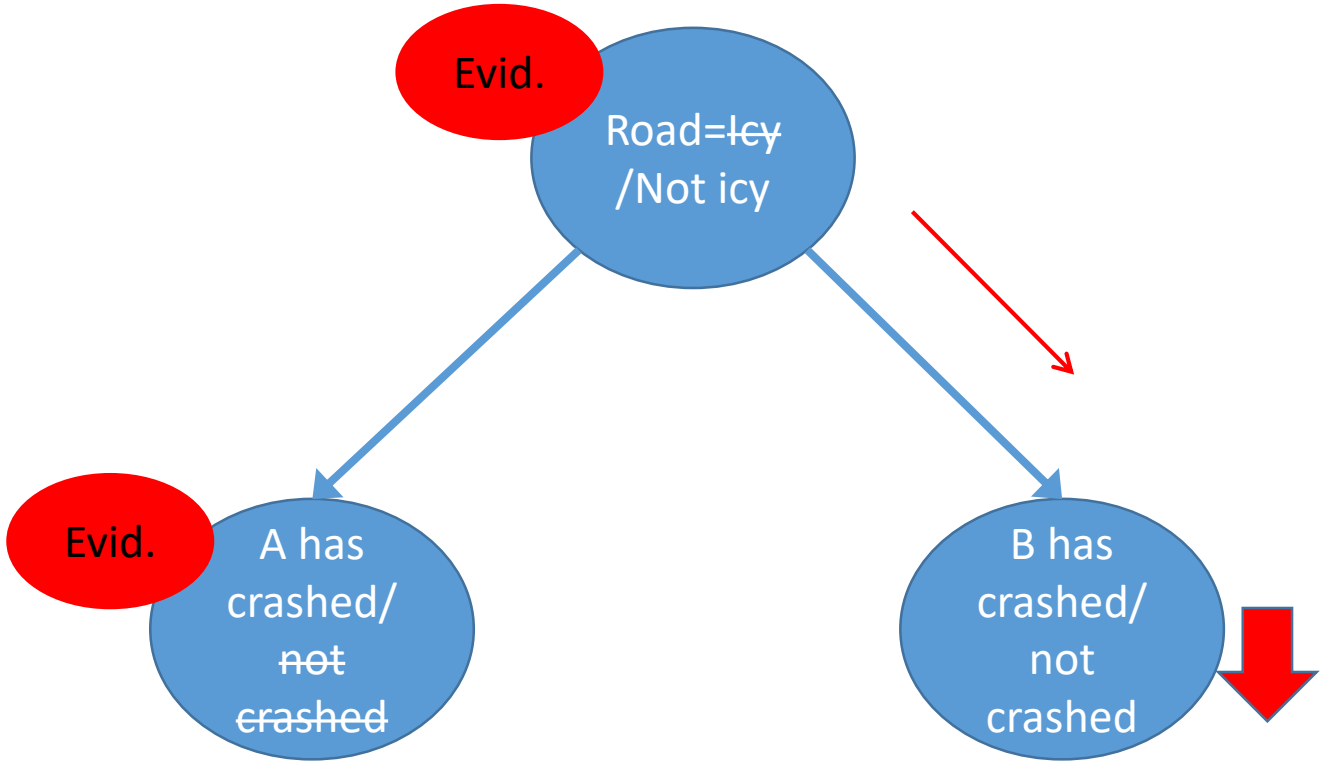
“Icy roads” example

- Initially, don’t know if road is icy, or if A/B had an accident
- You learn that A had an accident
  - What can we say about the condition of the road and B’s chances of an accident?
  - Our thinking:
    - “If A has crashed, then it probably means the roads are icy, and so it is more likely that B will have an accident”
  - The probability that B will have an accident **has increased given this information**


Road=Icy
/Not icy
Evidence
A has
crashed/
not
crashed
B has
crashed/
not
crashed

“Icy roads” example

- You now find out that roads are not icy
  - Our thinking:
    - “A was just unlucky; B will probably not have an accident”
  - The probability that B will have an accident **has decreased given this information**


Evid.
Road=~~Icy~~
/Not icy
Evid.
A has
crashed/
~~not~~
~~crashed~~
B has
crashed/
not
crashed

“Icy roads” joint distribution

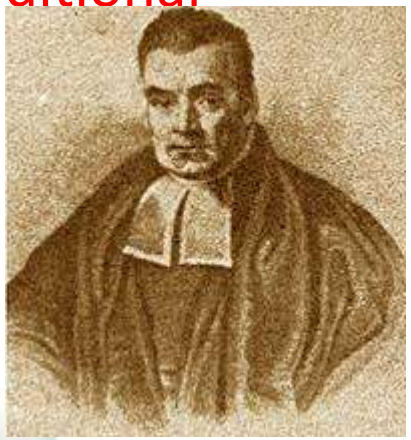
Road is icy?	A has accident?	B has accident?	P(Road is icy?, A has accident?, B has accident?)
False	False	False	0.648
False	False	True	0.072
False	True	False	0.072
False	True	True	0.008
True	False	False	0.098
True	False	True	0.042
True	True	False	0.042
True	True	True	0.018

Classwork: Compute

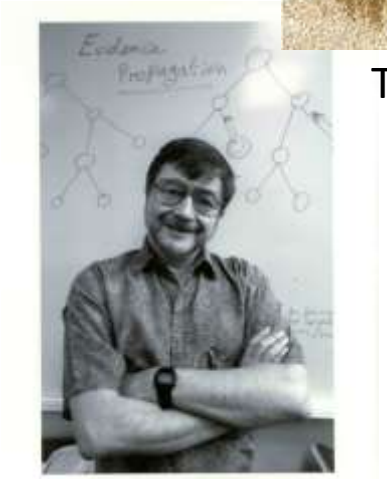
- (Prior) probability that A has accident
- Probability that A has accident if it is known that the road is icy
- Probability that A has accident if it is known that B had an accident (and it is not known if the road is icy or not)
- Probability that A has accident if it is known that B had an accident and Road is not icy

Bayesian Network Motivation

- We want a representation and reasoning system that is based on conditional independence
  - Compact yet expressive representation
  - Efficient reasoning procedures
- Bayesian Networks are such a representation
  - Named after Thomas Bayes (ca. 1702 –1761)
  - Term coined in 1985 by Judea Pearl (1936 –)
    - Turing Award winner, 2011
  - Invention changed the focus on AI from logic to probability!



Thomas Bayes



Judea Pearl

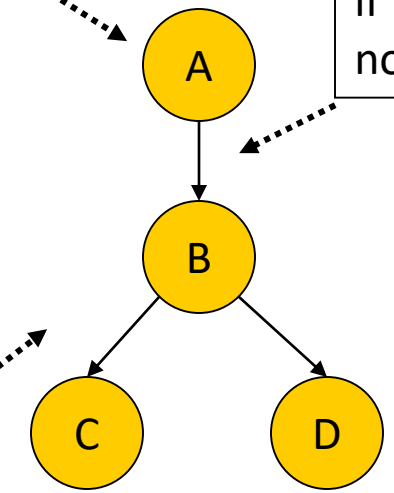
What are Bayesian networks?

- Bayesian networks (BN) are a graph-based framework for representing and analyzing models involving uncertainty
- Graph-based knowledge structure:
  - Nodes: variables
  - Edges: represent probabilistic dependence between variables
  - conditional probabilities encode the strength of the dependencies
- Algorithms:
  - computes posterior probabilities given evidence about some nodes
  - exploits probabilistic independence for efficient computation

Bayesian network: A Directed Acyclic Graph

Each node in the graph is a random variable

A node *X* is a parent of another node *Y* if there is an arrow from node *X* to node *Y* e.g. *A* is a parent of *B*



Informally, an arrow from node *X* to node *Y* means *X* has a direct influence on *Y*

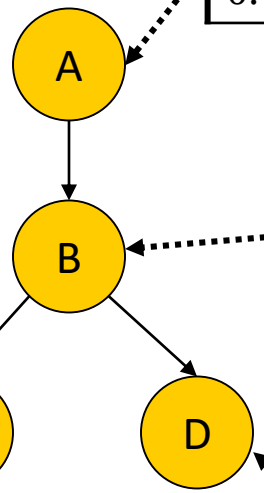
A Bayesian Network

1. A Directed Acyclic Graph
   - A node represents a variable
2. A conditional probability table for each node

Each node X_i has a conditional probability distribution $P(X_i \mid Parents(X_i))$ that quantifies the effect of the parents on the node

P(+a)	~~P(-a)~~
0.4	~~0.6~~

A	P(A)
false	0.6
true	0.4



A	P(+b|A)	~~P(-b|A)~~
false	0.99	~~0.01~~
true	0.3	~~0.7~~

A	B	P(B|A)
false	false	0.01
false	true	0.99
true	false	0.7
true	true	0.3

B	C	P(C|B)
false	false	0.4
false	true	0.6
true	false	0.9
true	true	0.1

B	P(+c|B)
false	0.6
true	0.1

B	P(+d|B)
false	0.98
true	0.95

B	D	P(D|B)
false	false	0.02
false	true	0.98
true	false	0.05
true	true	0.95

Examples of 3-way Bayesian Networks

P(+a)
0.4

P(+b)
0.75

P(+c)
0.33

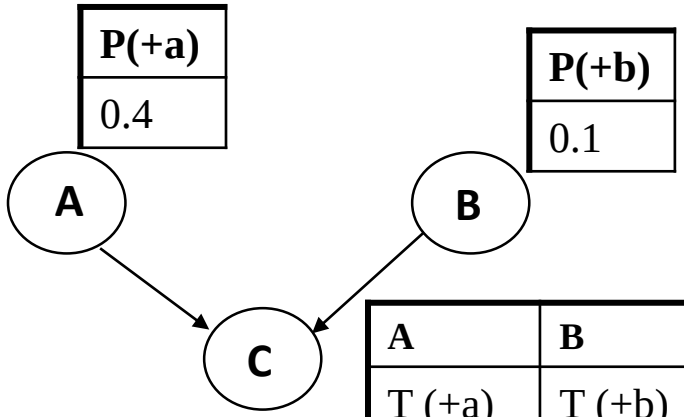
A B C

Absolute Independence:
p(A,B,C) = p(A) p(B) p(C)

- What do the 3 tables look like?
  - Assume all variables are binary

Examples of 3-way Bayesian Networks

- Independent Causes
  - A and B are independent
  - A and B **become dependent** once C is known
  - e.g., A is “exposure to toxins”, B is “exposure to radiation”
  - C is “cancer”



P(+a)
0.4

P(+b)
0.1

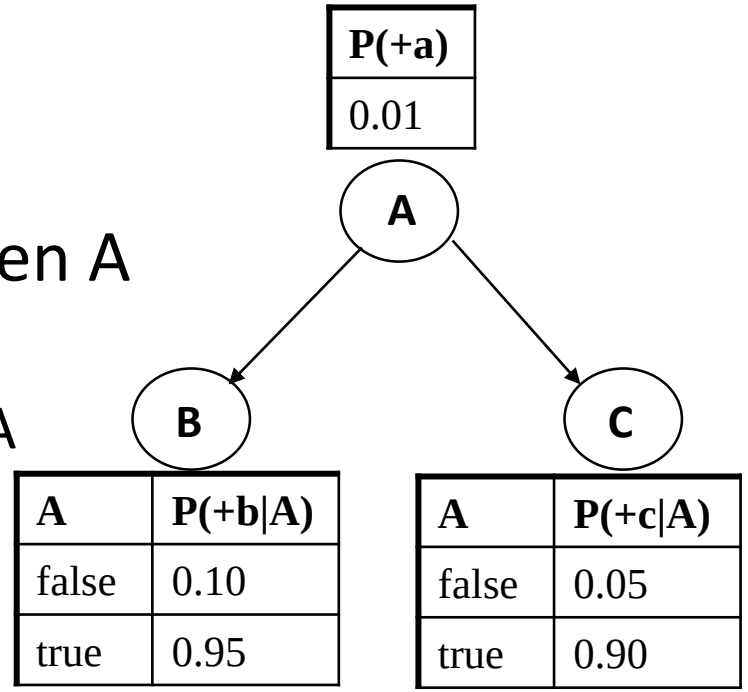
A	B	P(+c|A,B)
T (+a)	T (+b)	0.90
T	F	0.25
F (-a)	T	0.40
F	F	0.15

$$p(A,B,C) = p(A)p(B)p(C|A,B)$$

- What do the 3 tables look like?
  - Assume all variables are binary

Examples of 3-way Bayesian Networks

- Conditionally independent effects
- B and C are *not* independent
- But B and C **become conditionally independent** given A
  - e.g., A is a disease, and
  - B and C are conditionally independent symptoms given A
- $p(A, B, C) = p(A)p(B|A)p(C|A)$



P(+a)
0.01

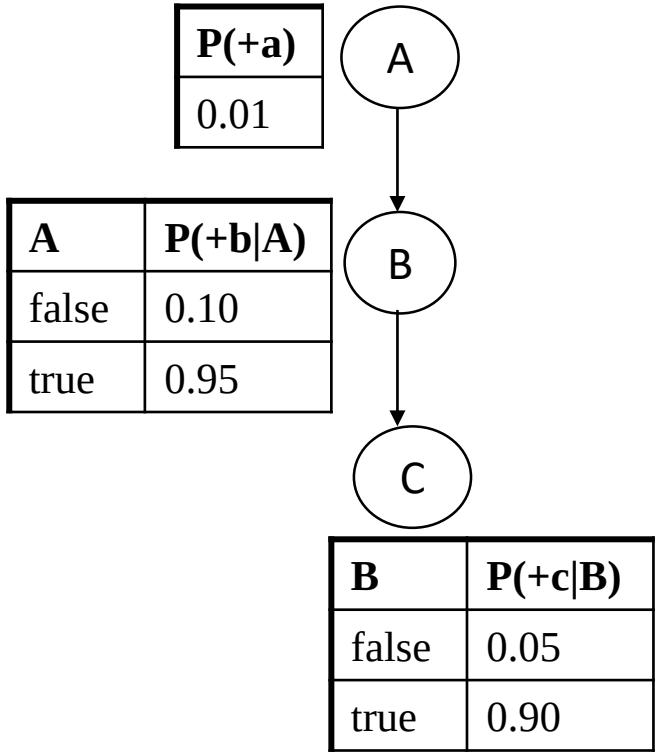
A	P(+b|A)
false	0.10
true	0.95

A	P(+c|A)
false	0.05
true	0.90

- What do the 3 tables look like?
  - Assume all variables are binary

Examples of 3-way Bayesian Networks

- Example: *causal chain*
- Are A and C independent?
- Is C independent of A given B?
- E.g.,
  - A: Low pressure
  - B: Rain
  - C: Traffic
- $p(A, B, C) = p(A)p(B|A)p(C|B)$



P(+a)
0.01

A	P(+b|A)
false	0.10
true	0.95

B	P(+c|B)
false	0.05
true	0.90

- What do the 3 tables look like?
  - Assume all variables are binary

The Joint Probability Distribution

Assuming conditional independence due to parents,

the joint probability distribution over all the variables X_1, …, X_n in the Bayesian network is:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^{n} P(X_i | Parents(X_i))$$

Compare to the general case:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^{n} P(X_i | X_1, \dots, X_{i-1})$$

Using a Bayesian Network Example

Using the network in the example, suppose you want to calculate:

P(A = true, B = true, C = true, D = true)

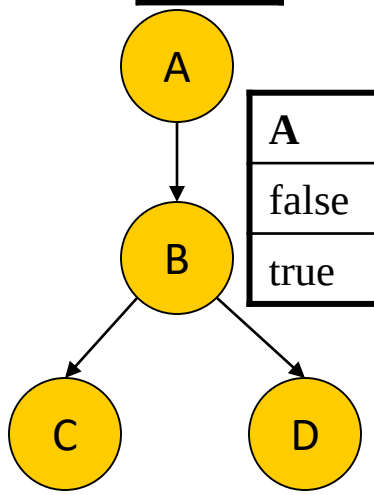
= P(A = true) * P(B = true | A = true) *

P(C = true | B = true) P(D = true | B = true)

= (0.4)*(0.3)*(0.1)*(0.95)

This is from the graph structure

These numbers are from the conditional probability tables



P(+a)
0.4

A	P(+b|A)
false	0.99
true	0.3

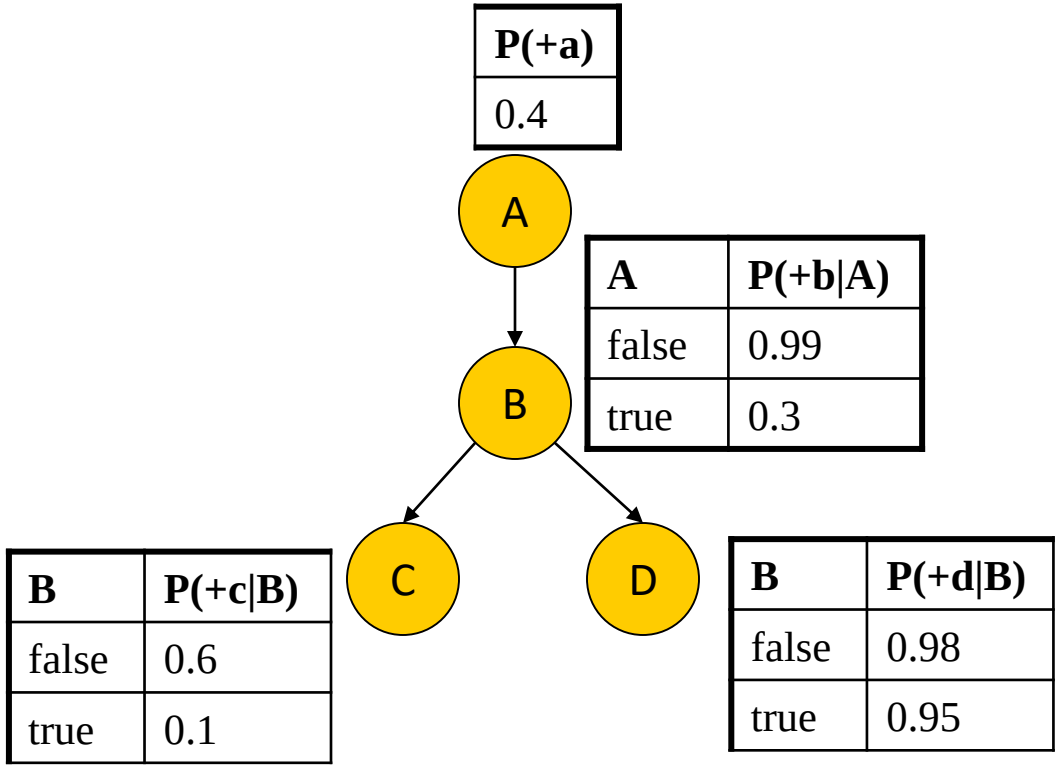
B	P(+c|B)
false	0.6
true	0.1

B	P(+d|B)
false	0.98
true	0.95

Classwork: Using a Bayesian Network

Using the network shown below, calculate:

P(A = true, B = false, C = false, D = true)



P(+a)
0.4

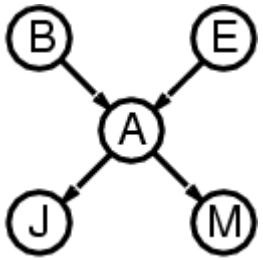
A	P(+b|A)
false	0.99
true	0.3

B	P(+c|B)
false	0.6
true	0.1

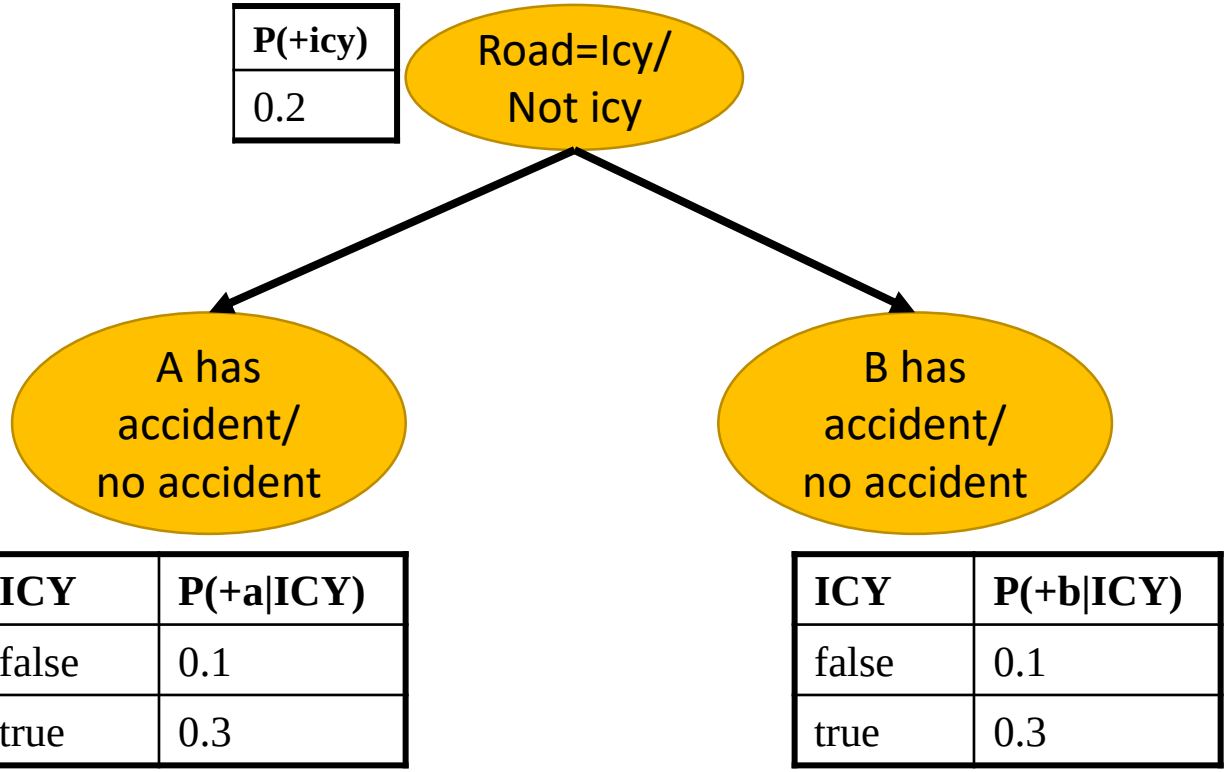
B	P(+d|B)
false	0.98
true	0.95

Compactness

- Suppose we have a Boolean variable X_i with k Boolean parents. How many rows does its conditional probability table have?
  - 2^k rows for all the combinations of parent values
  - Each row requires one number p for X_i = true
- If each variable has no more than k parents, how many numbers does the complete network require?
  - $O(n \cdot 2^k)$ numbers – vs. $O(2^n)$ for the full joint distribution



What is the joint distribution represented by this BN?



P(+icy)
0.2

ICY	P(+a|ICY)
false	0.1
true	0.3

ICY	P(+b|ICY)
false	0.1
true	0.3

What is the joint distribution represented by this BN?

Road is icy?	A has accident?	B has accident?	P(Road is icy?, A has accident?, B has accident?)
False	False	False	0.648
False	False	True	0.072
False	True	False	0.072
False	True	True	0.008
True	False	False	0.098
True	False	True	0.042
True	True	False	0.042
True	True	True	0.018

Inference

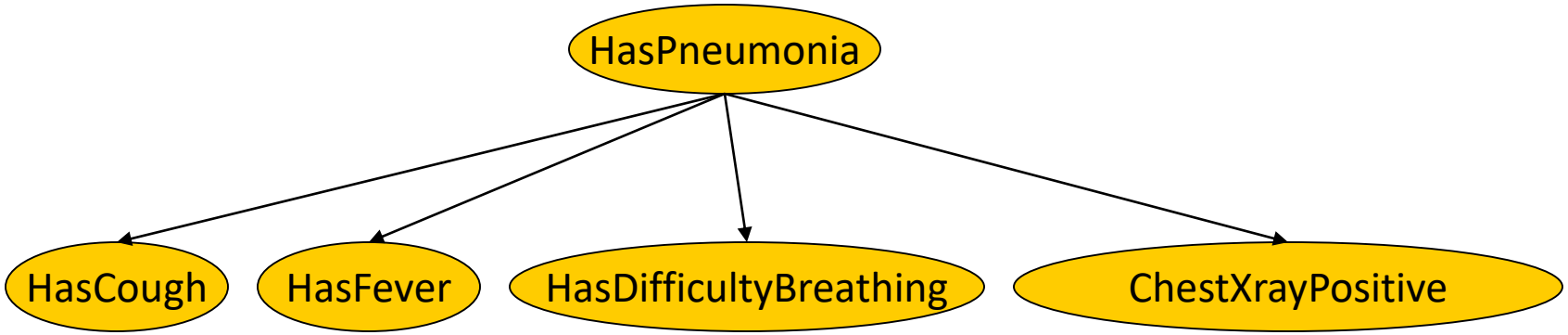
- Using a Bayesian network to compute probabilities is called inference
- In general, inference involves queries of the form:

$P(X \mid E)$

E = The evidence variable(s)

X = The query variable(s)

Inference

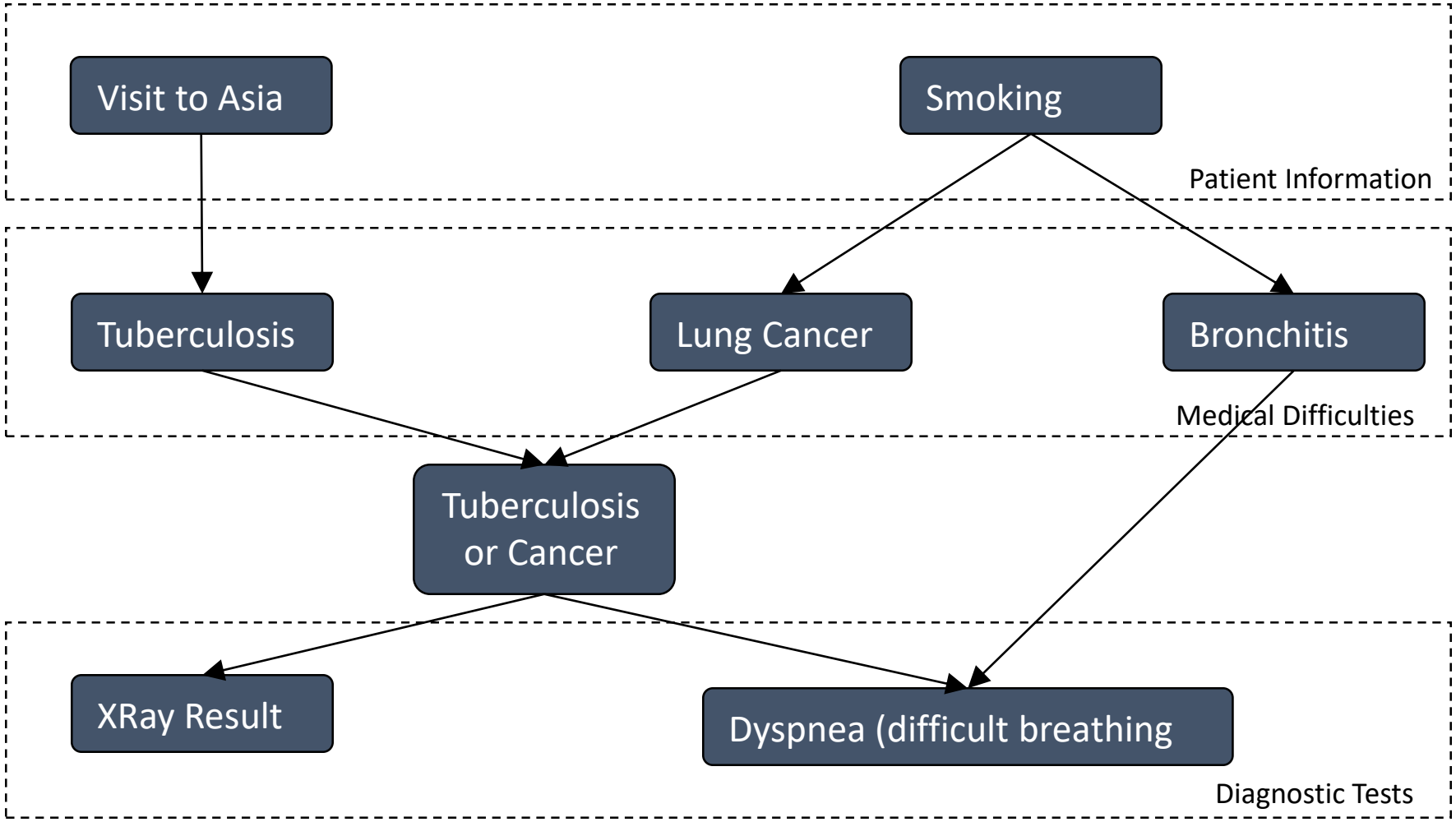


- An example query:

  P(*HasPneumonia = true* | *HasFever = true*, *HasCough = true*)?

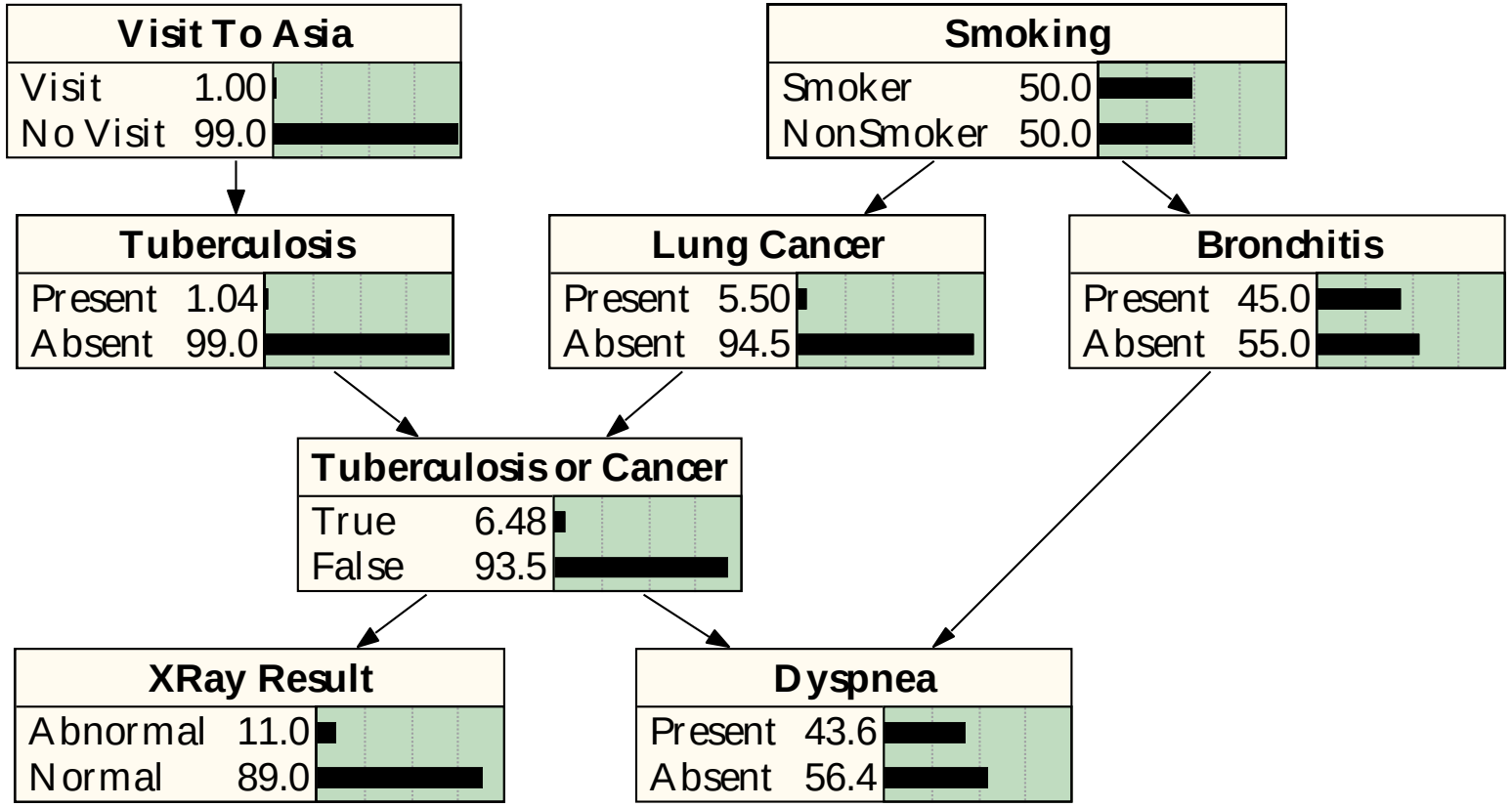
- Note: Even though *HasDifficultyBreathing* and *ChestXrayPositive* are in the Bayesian network, they are not given values in the query
  - Neither query variables nor evidence variables
  - They are treated as unobserved or hidden variables

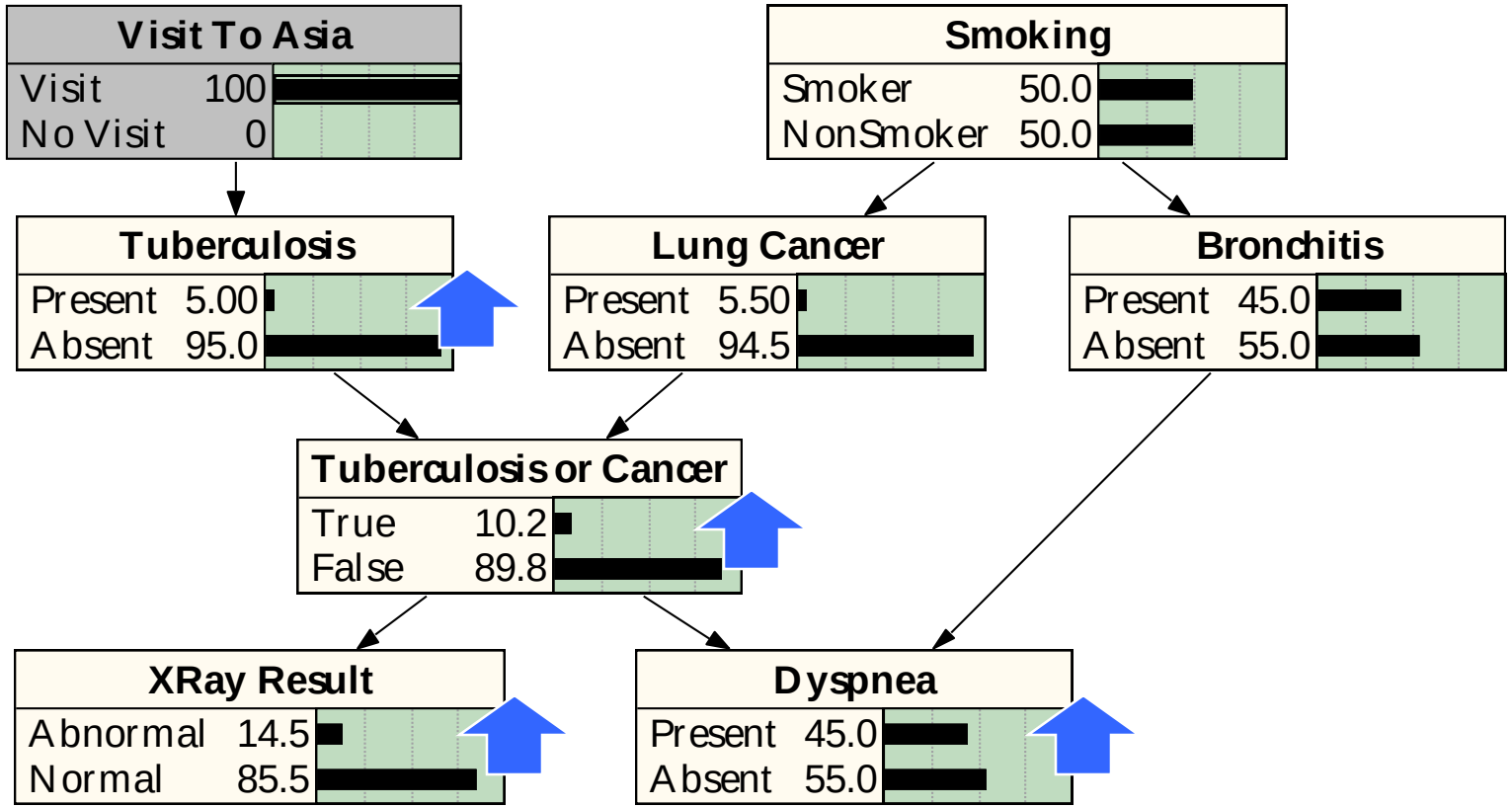
Example from Medical Diagnostics



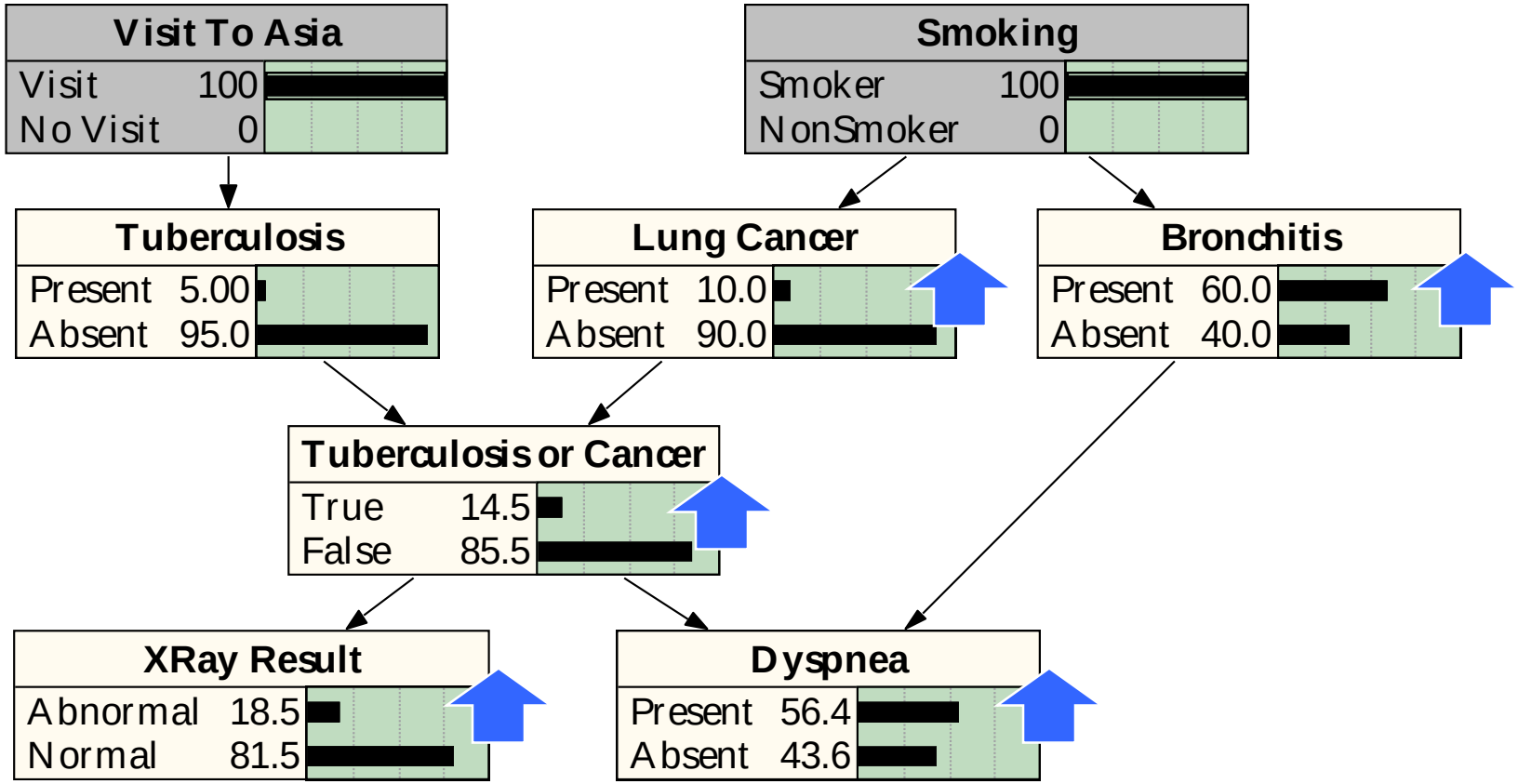
- Network represents a knowledge structure that models the relationship between medical difficulties, their causes and effects, patient information and diagnostic tests

From D.M. Buede, J.A. Tatman, T.A. Bresnick "Introduction to Bayesian Networks," Tutorial for the 66th MORS Symposium

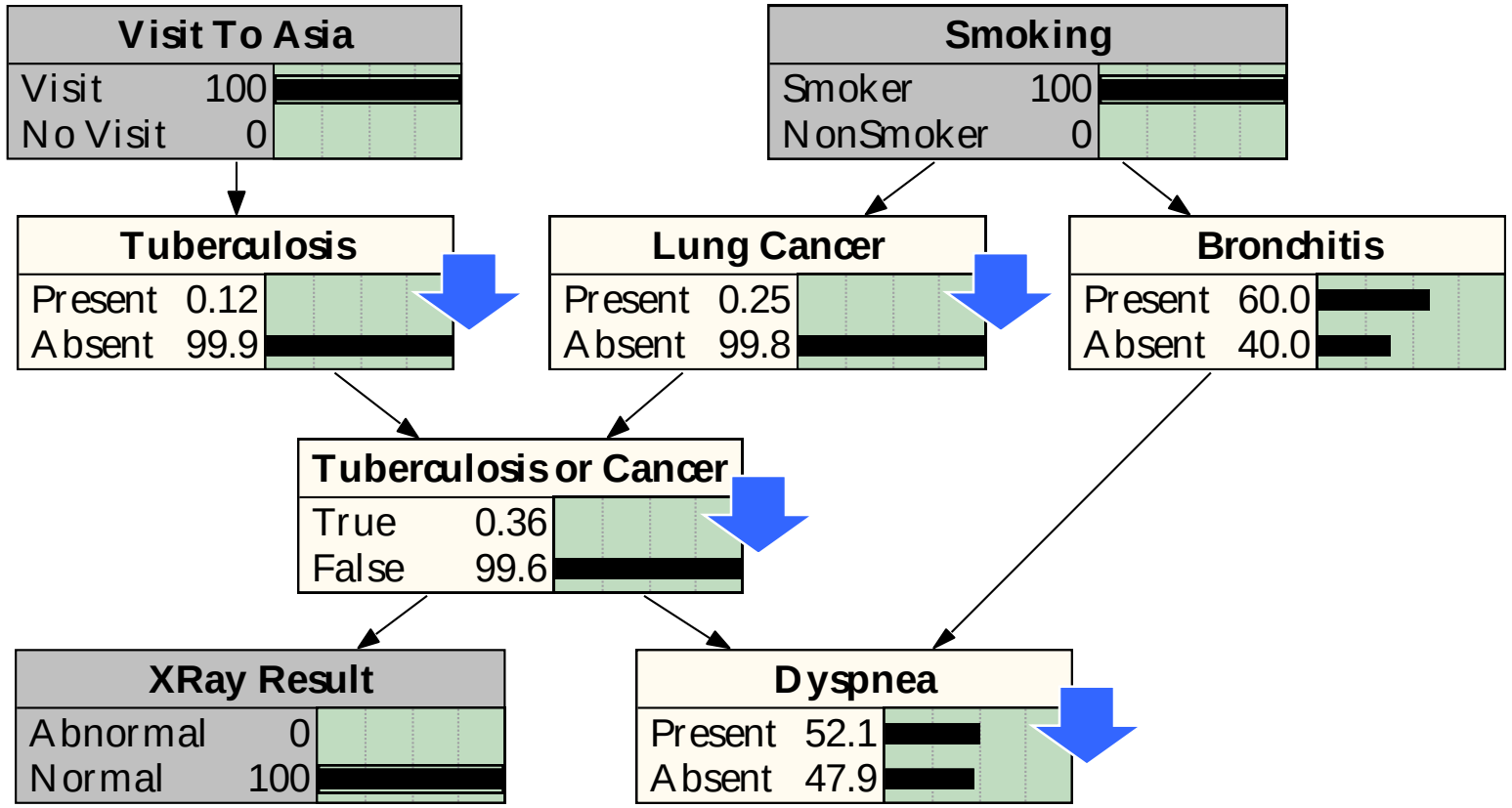

Visit To Asia
Visit 1.00
No Visit 99.0
Smoking
Smoker 50.0
NonSmoker 50.0
Tuberculosis
Present 1.04
Absent 99.0
Lung Cancer
Present 5.50
Absent 94.5
Bronchitis
Present 45.0
Absent 55.0
Tuberculosis or Cancer
True 6.48
False 93.5
XRay Result
Abnormal 11.0
Normal 89.0
Dyspnea
Present 43.6
Absent 56.4



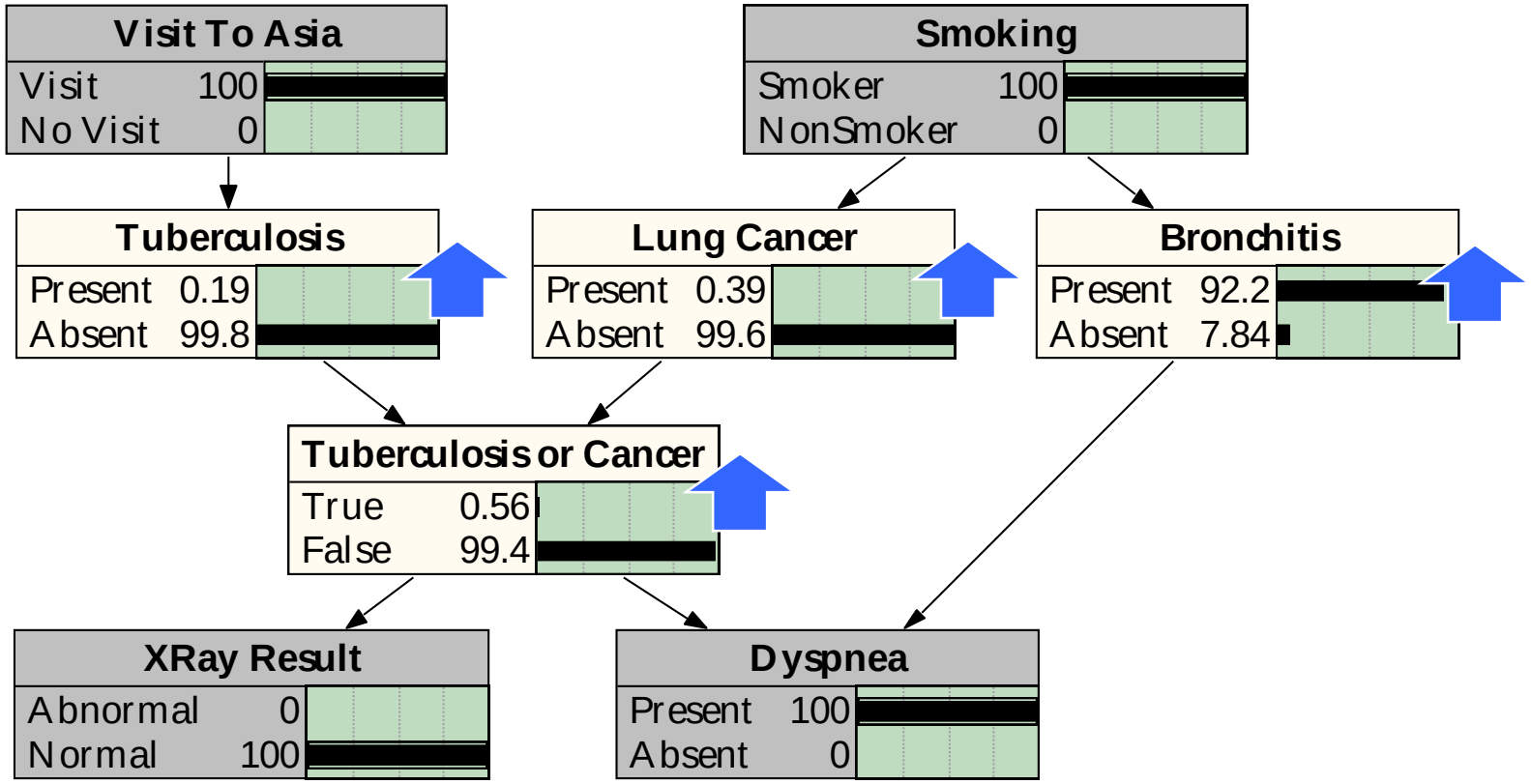
- As a finding is entered, the propagation algorithm updates the beliefs attached to each relevant node in the network
- Interviewing the patient produces the information that “Visit to Asia” is “Visit”
- This finding propagates through the network and the belief functions of several nodes are updated



- Further interviewing of the patient produces the finding “Smoking” is “Smoker”
- This information propagates through the network



- Finished with interviewing the patient, the physician begins the examination
- The physician now moves to specific diagnostic tests such as an X-Ray, which results in a “Normal” finding which propagates through the network
- Note that the information from this finding propagates backward and forward through the arcs

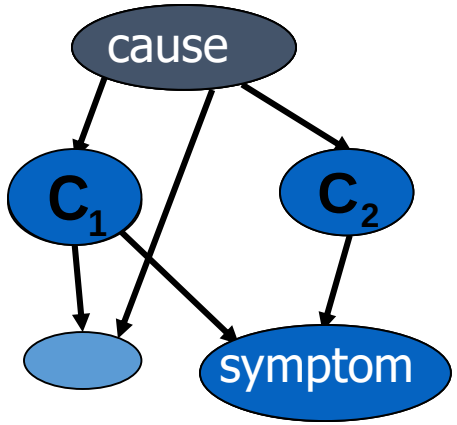


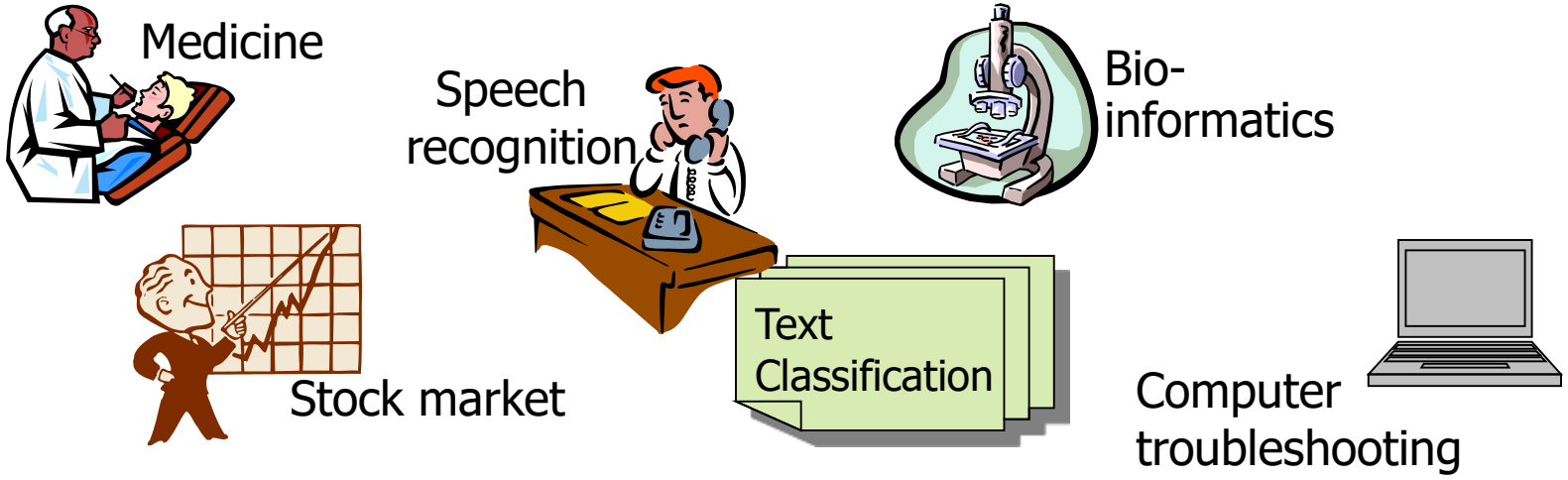
- The physician also determines that the patient is having difficulty breathing, the finding "Present" is entered for "Dyspnea" and is propagated through the network
- The doctor might now conclude that the patient has bronchitis and does not have tuberculosis or lung cancer

Applications of BNs

- Diagnosis: P(cause|symptom)=?
- Prediction: P(symptom|cause)=?

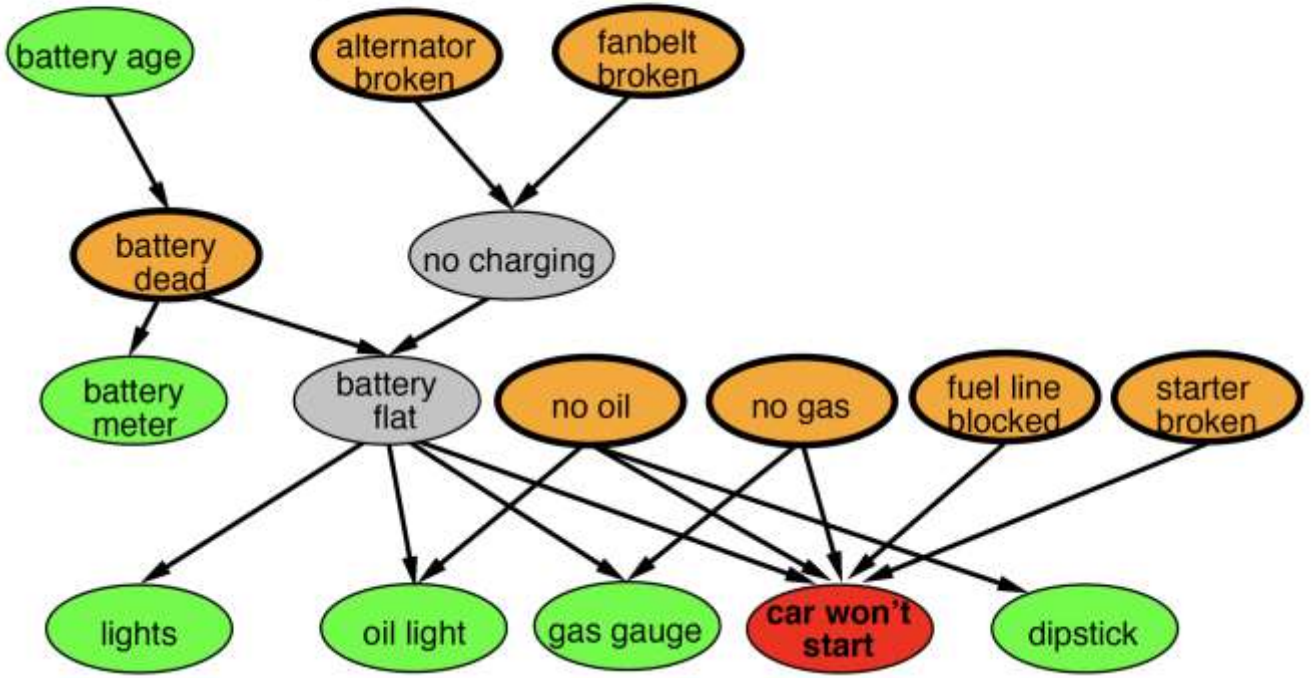
- Classification: $\max_{class}$ P(class|data)
- Decision-making (given a cost function)





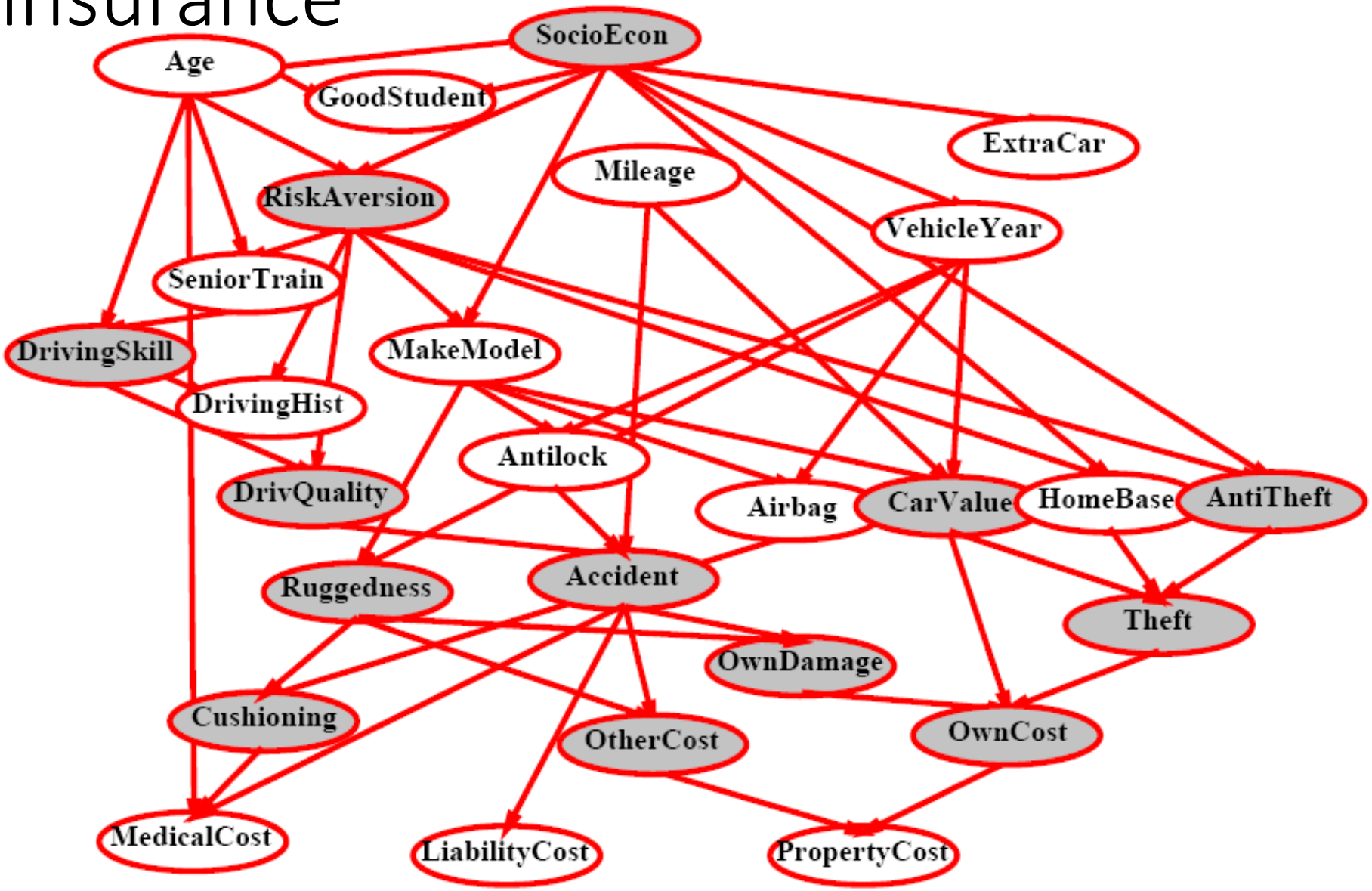
Application: car diagnosis

Initial evidence: car won't start
Testable variables (green), "broken, so fix it" variables (orange)
Hidden variables (gray) ensure sparse structure, reduce parameters

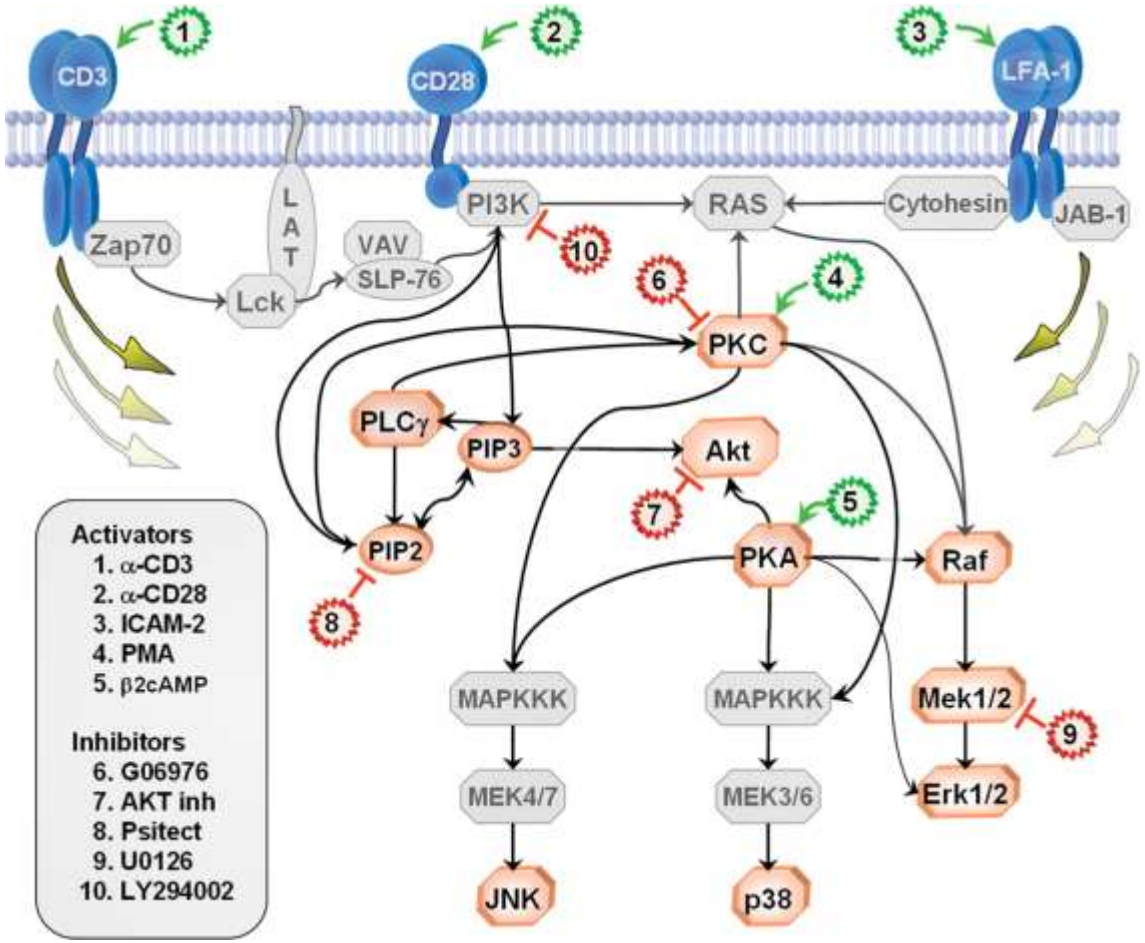


Russel and Norvig

Car insurance



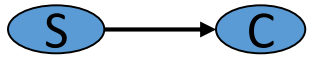
In research literature…

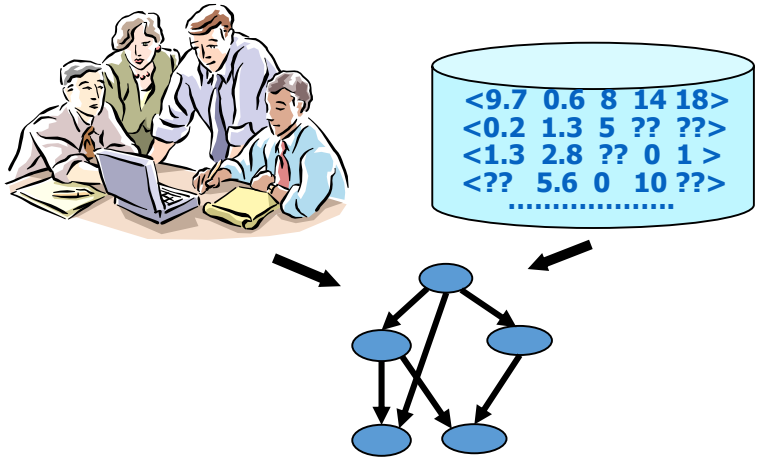


Causal Protein-Signaling Networks Derived from Multiparameter Single-Cell Data

Karen Sachs, Omar Perez, Dana Pe'er, Douglas A. Lauffenburger, and Garry P. Nolan (22 April 2005) *Science* **308** (5721), 523.

Advantages of BNs

- Combining domain expert knowledge with data
- Efficient representation and inference
- Handling missing data: Not all variable states need to be known in a query
- Learning causal relationships: 



Why does this work?

The Chain Rule

- The product rule

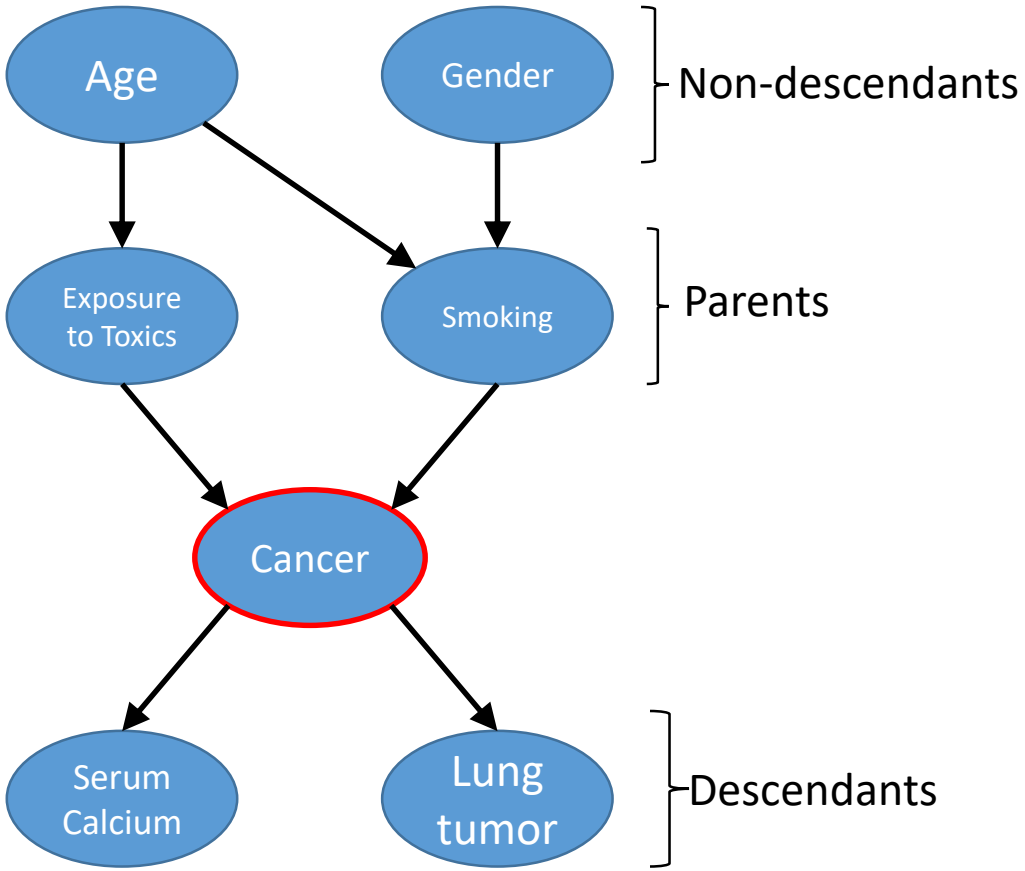
$$P(X_1, X_2) = P(X_1)P(X_2|X_1)$$

- More generally, can always write **any** joint distribution as an incremental product of conditional distributions
- Arrange the variables in ***any*** order: $X_1, X_2, \dots, X_n$
  - Condition each variable on ***all*** the variables that appear before it

$$P(X_1, X_2, X_3) = P(X_1)P(X_2|X_1)P(X_3|X_1, X_2)$$

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^{n} P(X_i|X_1, \dots, X_{i-1})$$

Conditional independence



A variable (node) is **conditionally independent** of its non-descendants given *all* its parents

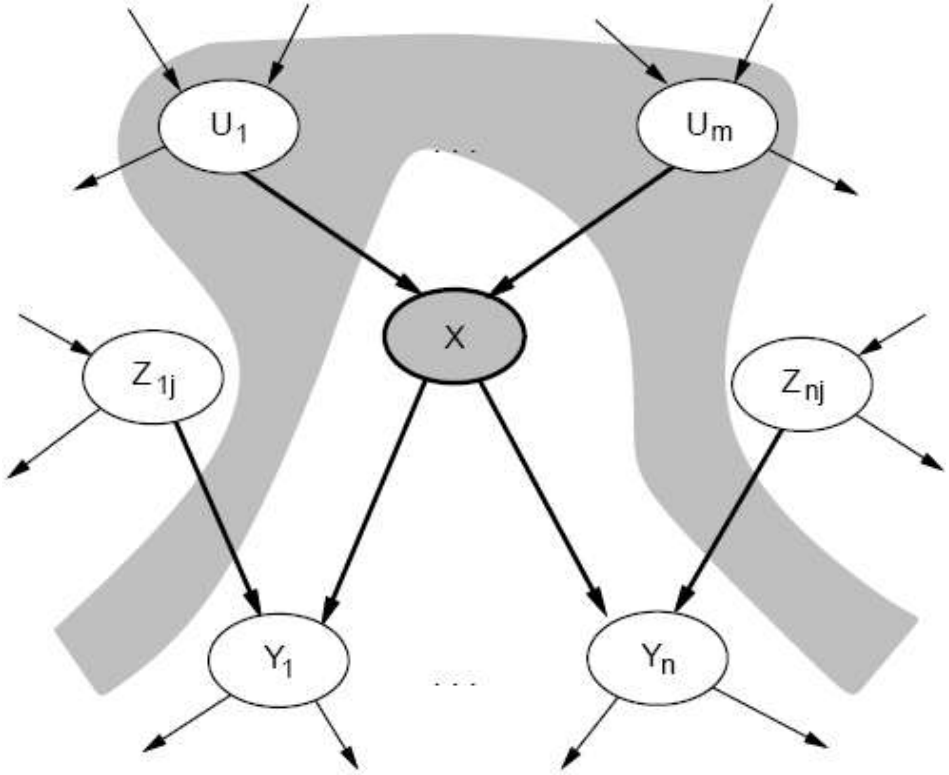
P is conditionally independent of Q if
(1) Q is a non-descendant of P
(2) given *all* parents(P)

Cancer is independent of *Age* and *Gender* given *Exposure to Toxics* and *Smoking*

- *Age* and *Gender* are non-descendants of *Cancer*
- *Exposure to Toxics* and *Smoking* are all the parents of *Cancer*

Conditional Independence

A node (X) is conditionally independent of its non-descendants (Z_{1j}, Z_{nj}), given its parents (U_1, U_m).



The Joint Probability Distribution

Assuming conditional independence due to parents,

the joint probability distribution over all the variables X_1, …, X_n in the Bayesian network is:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^{n} P(X_i | Parents(X_i))$$

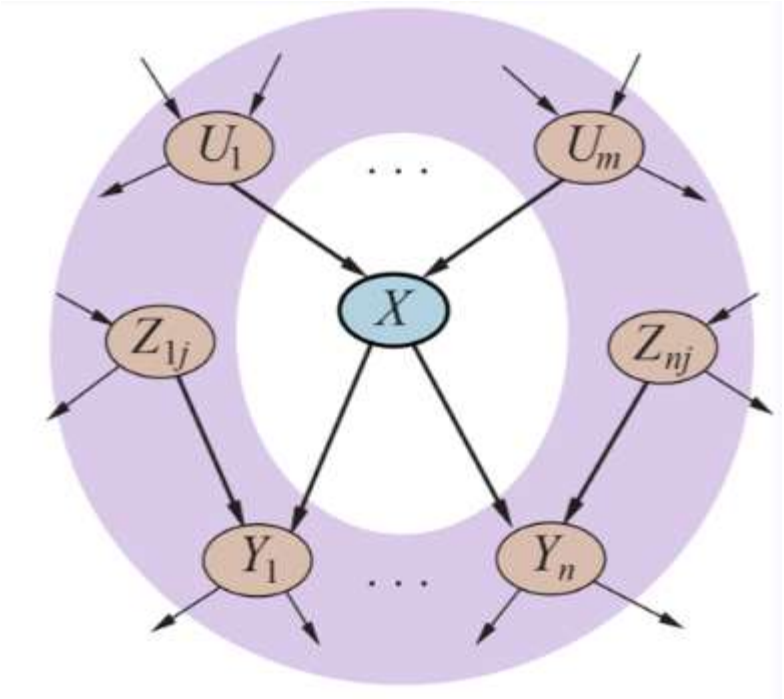
Compare to the general case:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^{n} P(X_i | X_1, \dots, X_{i-1})$$

Markov blanket

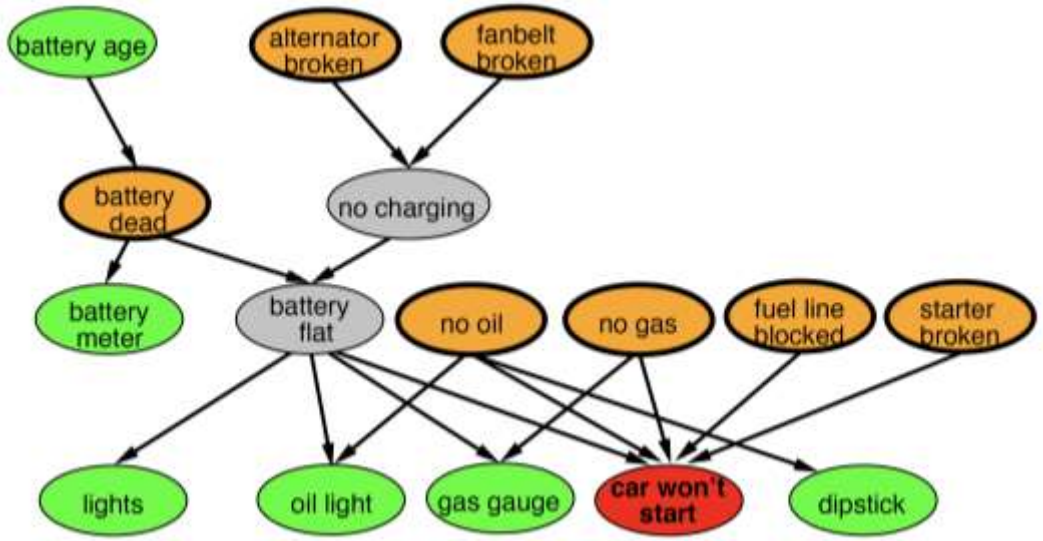
- A variable is conditionally independent of **all** other nodes, given all its:
  - parents
  - children
  - children's parents

  Markov blanket of the variable



Class work

- What is the Markov blanket of
  1. no charging
  2. battery dead



Inference in Bayesian Networks

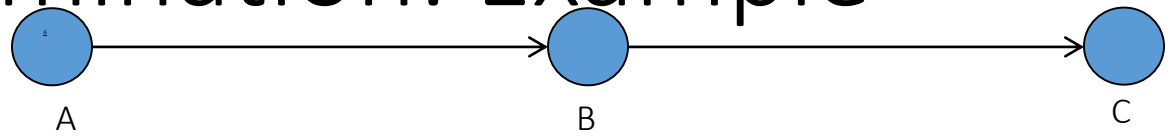
Exact Inference in Bayesian networks

- The general probabilistic inference problem is to find the probability of an event given a set of evidence
- In theory:
  - Just construct the entire joint distribution and then marginalize (to remove variables not in the query)
  - Exponential in the number of variables
    - The full joint distribution includes every possible combination of variables
    - In time and space
- In practice:
  - We need more efficient algorithms
  - Idea: no need to create the full joint distribution and then marginalize
  - Do marginalization (remove variables) *before* creating the joint distribution
  - Fewer variables -> smaller table
- Variable Elimination algorithm

Variable Elimination algorithm

- An exact inference algorithm for BNs
- Worst-case exponential complexity, but practical for certain BNs
  - With low “tree-width”: informally, how far is the graph is from being a tree?
- Intuitive idea:
- Similar to simplification of algebraic formula with multiplication and addition
- a*b + a*c + a*d + a*e*h + a*f*h + a*g*h.
  - It takes 14 multiplications or additions to evaluate the expression
- a (b + c + d + h*(e + f + g))
  - This expression only needs 7 operations

Variable Elimination: Example



- We want to compute $P(C)$

$$P(C) = \sum_A \sum_B P(A,B,C) = \sum_A \sum_B P(A)P(B|A)P(C|B)$$

$$= \sum_B P(C|B) \boxed{\sum_A P(A)P(B|A)}$$

Let's call this $f_A(B)$

$$= \sum_B P(C|B) f_A(B)$$

A has been (instantiated and) eliminated

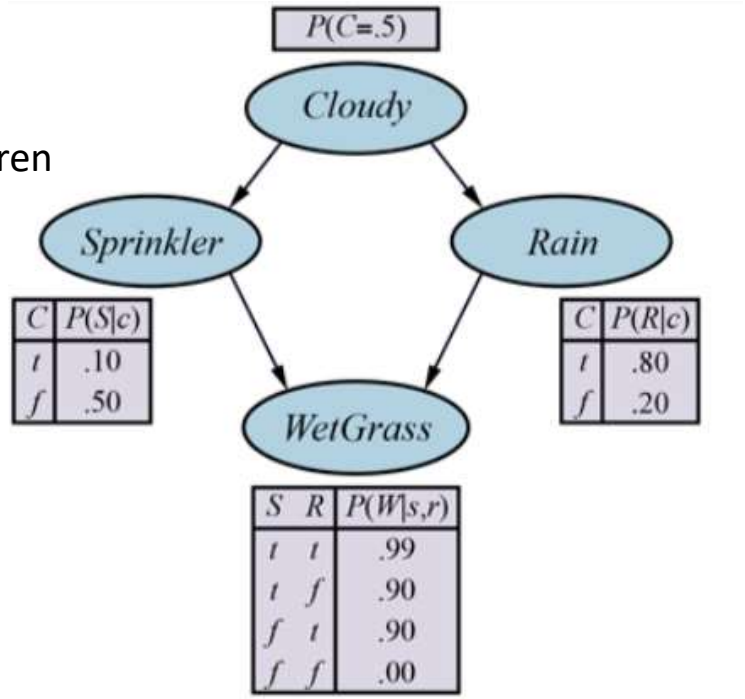
- Why is this better?
  - Fewer multiplications and additions

Approximate Inference

- Even the most efficient exact inference algorithms are too slow on large complex networks
- In these cases, we can approximate inferences by sampling.
  - Sampling is a technique to select and count the occurrences of the query and evidence variables to estimate the probability distributions in the network.
  - Stochastic approach to calculate approximate answers to a query
- Typical query: $p(x|E)$?
  - E: set of evidence variables with *known values*
  - X: subset of variables whose probabilities I want to know
  - $p(x|E)$ is called the *posterior distribution*
- Core idea:
  - Set the values of evidence variables to known values
  - *randomly* generate samples from the Bayesian network
  - Keep track of how many samples agree with x

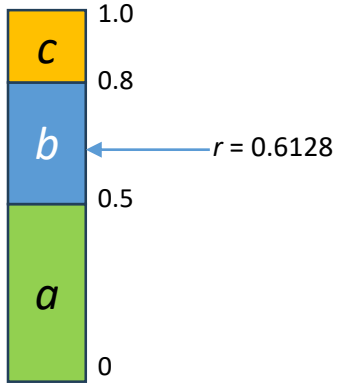
Direct sampling

- for i=1:N do:
  - Sample each variable one-at-a-time, in topological order
    - **Topological order**: any order of variables, where parents appear before children
      - E.g.: Cloudy, Sprinkler, Rain, WetGrass
      - E.g.: Cloudy, Rain, Sprinkler, WetGrass
  - Sample variable X_i from $P(X_i|Parents(X_i))$
    - Note: values of parents are known at this time (because of topological order)
- Repeat many times (N is a large number)
- We get N events
  1. Cloudy=*t*, Sprinkler=*f*, Rain=*t*, WetGrass=*f*
  2. Cloudy=*t*, Sprinkler=*t*, Rain=*t*, WetGrass=*f*
  3. …
- Count how many of these match the specific event of interest ($N_{specific}$)
  - Answer: $P(specific\ event) = \frac{N_{specific}}{N}$



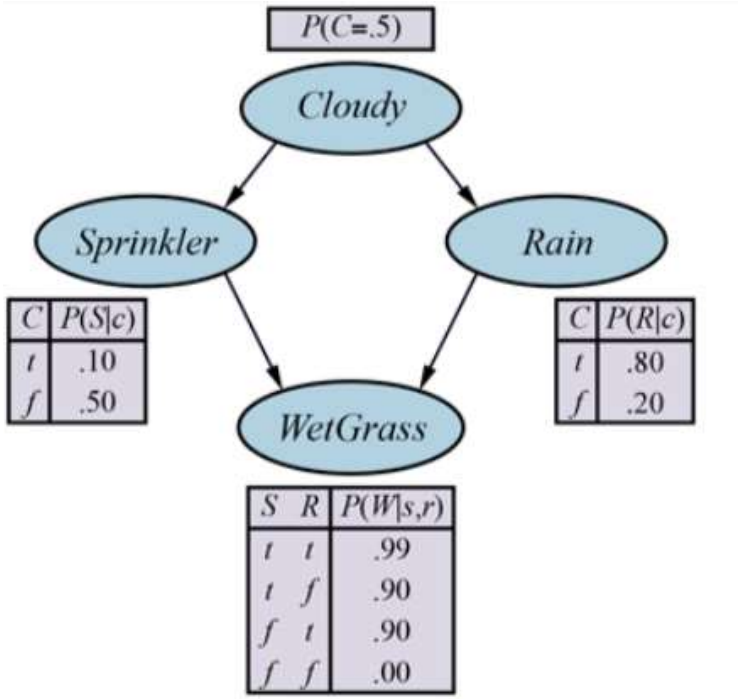
Sampling a variable

- *Sampling*: Randomly setting **the value** of a variable from a distribution
- How? Generate a random number and see where it falls on its *cumulative distribution*
- Example:
  - X can take three values: a, b, c
  - Sample X from probability distribution $\langle 0.5, 0.3, 0.2 \rangle$
  - Calculate cumulative distribution $\langle 0.5, 0.8, 1.0 \rangle$
  - $r \leftarrow$ generate a random number between 0 and 1
    - If $0 < r < 0.5$: $\mathrm{X} = a$
    - If $0.5 < r < 0.8$: $\mathrm{X} = b$
    - If $0.8 < r < 1.0$: $\mathrm{X} = c$



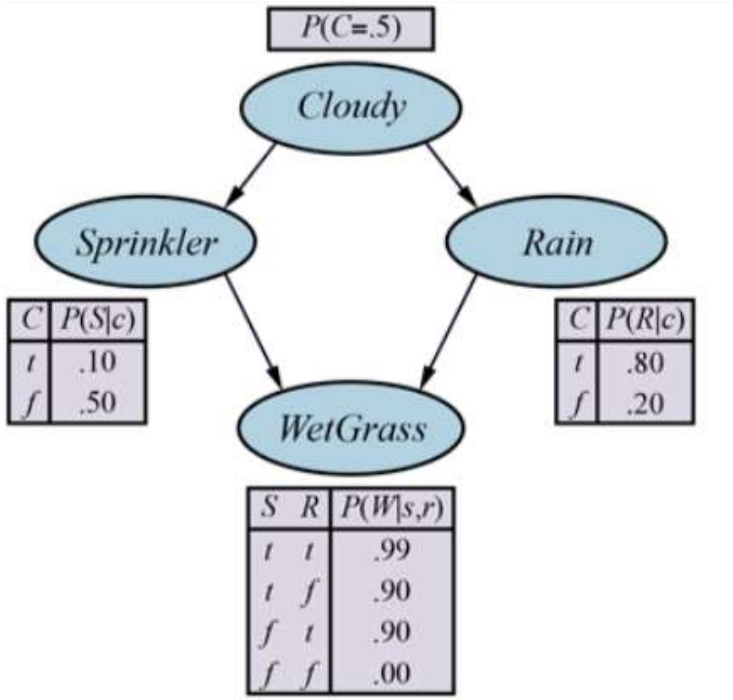
Direct sampling example

- Topological order:
  - Cloudy, Sprinkler, Rain, WetGrass
- One iteration of random sampling
  1. Sample from $P(Cloudy) = \langle 0.5, 0.5 \rangle$, value=true
  2. Sample from $P(Sprinkler|Cloudy) = \langle 0.1, 0.9 \rangle$, value=false
  3. Sample from $P(Rain|Cloudy) = \langle 0.8, 0.2 \rangle$, value=true
  4. Sample from $P(WetGrass|Sprinkler, Rain) = \langle 0.9, 0.1 \rangle$, value=true
- Event:
  - Cloudy, Sprinkler, Rain, WetGrass = (true, false, true, true)



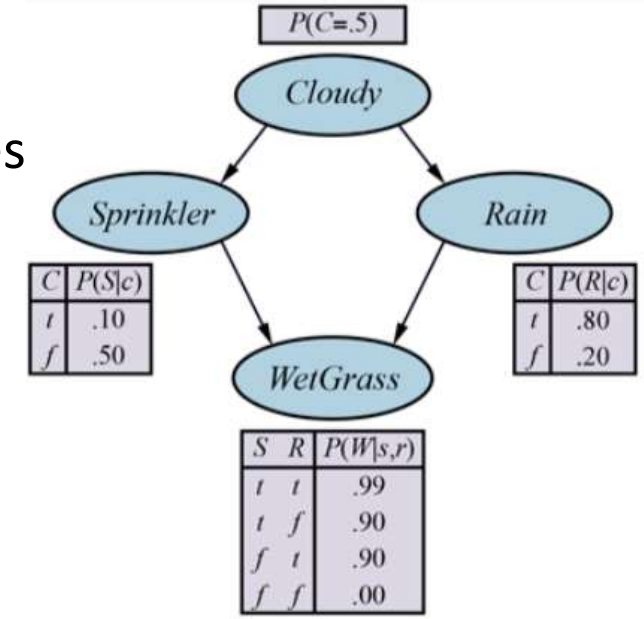
Markov Chain Monte Carlo (MCMC) method

- *Core idea*: instead of generating each sample from scratch, generate a sample by making a single variable change to the previous sample
- *Current state*: a value for every variable
- *Next state*: make a random change to one variable in the current state
- *Markov chain*: a random process that generate a sequence of states
  - Perform a *random walk* on this Markov chain
- Gibbs Sampling for Bayesian Networks
  - Initialize Current state = random assignment of values to all variables, except Evidence variables (which have already known values)
  - Iterate many times:
    - Choose one (non-evidence) variable of current state (X_i)
    - Sample X_i from $P(X_i|MarkovBlanket(X_i))$
    - Keep track of value counts for X_i



Cloudy	Sprinkler	Rain	WetGrass
true	**true**	false	**true**
false	**true**	false	**true**
...	...	...	...

Gibbs Sampling for Bayesian Networks



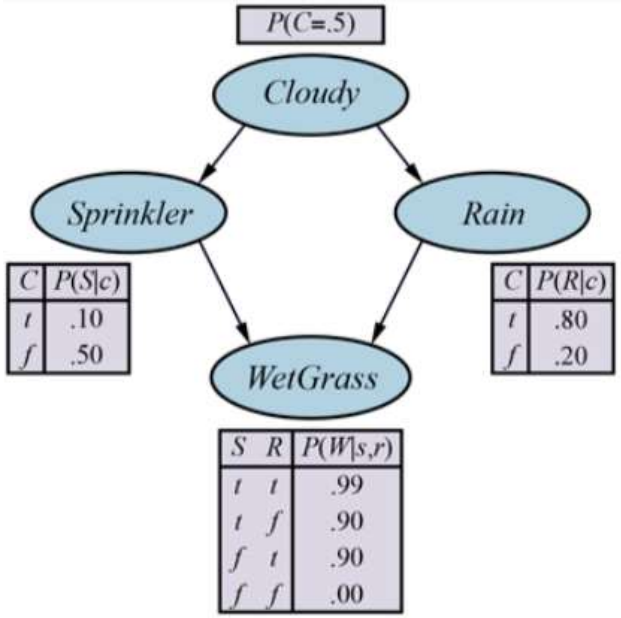
- Initialize Current state = random assignment of values to all variables
  - **except** Evidence variables (which have already known values)
- Iterate many times:
  - Choose one (non-evidence) variable of current state (X_i)
  - Sample X_i from $P(X_i|MarkovBlanket(X_i))$
  - Keep track of value counts for all X_i you are interested in
- Final answer: normalize value counts to 1.0
- $P(X_i|MarkovBlanket(X_i)) = \alpha P(X_i|Parents(X_i)) \prod_{Y \in Children(X_i)} P(y|Parents(Y))$
- i.e., multiply the probabilities in the tables of X_i

  and all its children

Cloudy	Sprinkler	Rain	WetGrass
true	**true**	false	**true**
false	**true**	false	**true**
...	...	...	...

Gibbs Sampling example

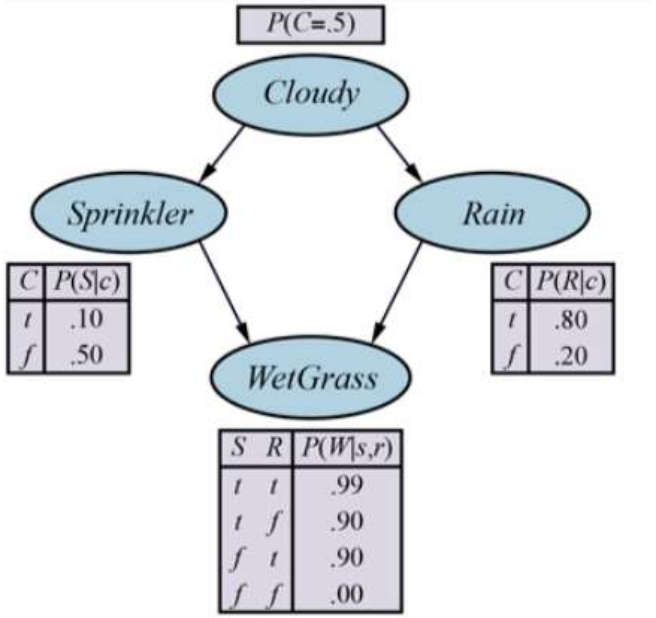
- To estimate $P(Rain|Sprinkler = true, WetGrass = true)$
  - Four variables
  - Two are evidence variables (Sprinkler and WetGrass) with known values (always true)
- Initialize Current state = (true, **true**, false, **true**)
- First iteration:
  - (Randomly) select Cloudy
  - Sample Cloudy's value from $P(Cloudy|Sprinkler = true, Rain = false)$
  - $P(c| + s, -r) = \alpha \times 0.5 \times 0.1 \times 0.2 = \alpha 0.001$
  - $P(-c| + s, -r) = \alpha \times 0.5 \times 0.5 \times 0.8 = \alpha 0.020$
  - So sampling distribution for Cloudy is $\left\langle \frac{0.001}{0.001+0.020}, \frac{0.020}{0.001+0.020} \right\rangle = \langle 0.048, 0.952 \rangle$
  - Say random value is Cloudy=false
  - Next state = (false, **true**, false, **true**)
  - Count[Rain=false]++
- Class work: second iteration
- After 1000 iterations, if Count[Rain=true]=200, and Count[Rain=false]=800
  - $P(Rain|Sprinkler = true, WetGrass = true)$ = 200/(200+800)=0.2

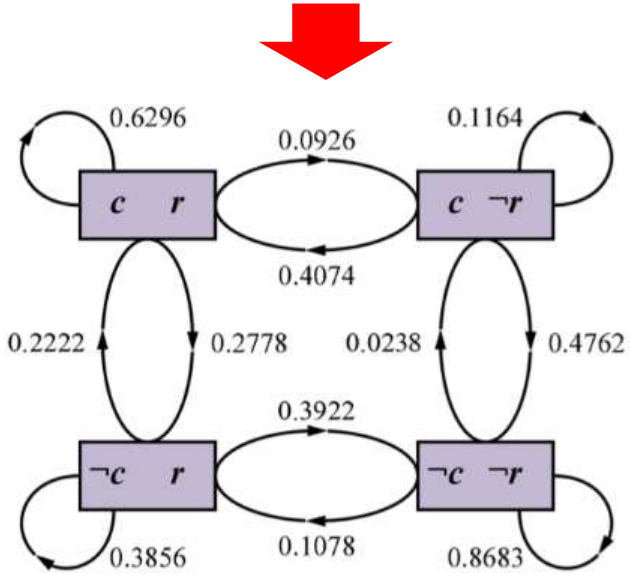


Cloudy	Sprinkler	Rain	WetGrass
true	**true**	false	**true**
false	**true**	false	**true**
...	...	...	...

Why does Gibbs Sampling work?

- The algorithm moves from state to state following certain transition probabilities
  - A Random Walk around the Markov Chain
- "Searching" for states with relatively high probabilities
- The Markov chain approximates the Bayesian Network's probability distribution
  - Markov chain's equilibrium distribution matches the true distribution
- The more steps that are "walked", the more closely the distribution of the samples matches the true distribution.





Constructing Bayesian Networks

How is the Bayesian network created?

1. Get an expert to design it
   - Expert must determine the structure of the Bayesian network
     - This is best done by modeling direct causes of a variable as its parents
   - Expert must determine the values of the CPT entries
     - These values could come from the expert's informed opinion
     - Or an external source e.g. census information
     - Or they are estimated from data
     - Or a combination of the above

2. **Learn** it from data
   - This is a much better option but it requires a large amount of data

Constructing Bayesian networks

1. Choose an ordering of variables $X_1, \ldots, X_n$
2. For i = 1 to n
   - add X_i to the network
   - select parents from $X_1, \ldots, X_{i-1}$ such that
     $P(X_i \mid Parents(X_i)) = P(X_i \mid X_1, \ldots X_{i-1})$

Example for BN construction: Fire Diagnosis

You want to diagnose whether there is a fire in a building

- If there is a fire, there *may* be smoke
- If there is a fire alarm, it *may* have been caused by a fire *or* by tampering
- If everyone is leaving, this *may* have been caused by a fire alarm
- You receive a *noisy* report about whether everyone is leaving the building

Example for BN construction: Fire Diagnosis

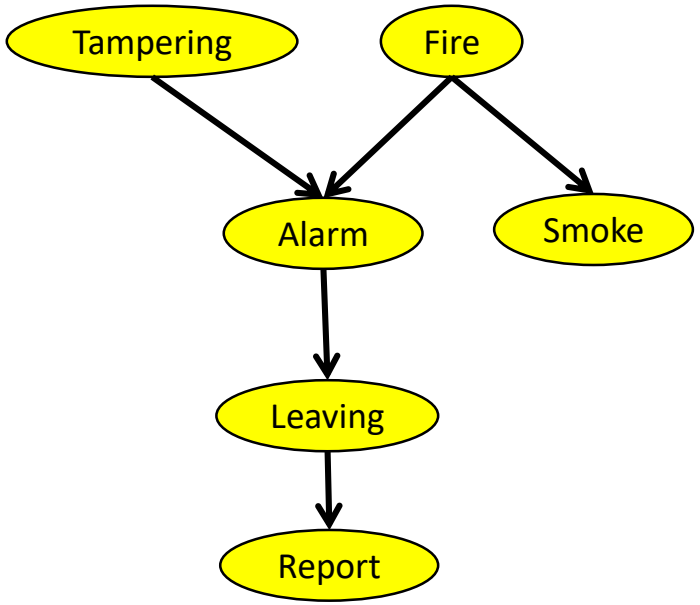
First you choose the variables. In this case, all are Boolean:

- Fire is true when there is a fire
- Smoke is true when there is smoke
- Alarm is true when there is an alarm
- Tampering is true when the alarm has been tampered with
- Leaving is true if there are lots of people leaving the building
- Report is true if the news reports that lots of people are leaving the building

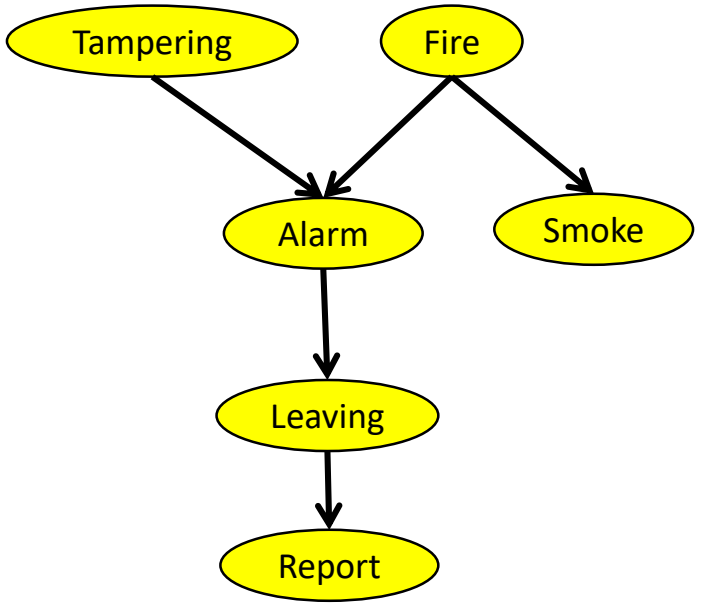
- Let’s construct the Bayesian network for this

Example for BN construction: Fire Diagnosis

- Using the total ordering of variables:
  - (1)Fire, (2) Tampering, (3) Alarm, (4) Smoke, (5) Leaving, (6) Report
- Choose the parents for each variable by evaluating conditional independencies
  - Fire is the first variable in the ordering. It does not have parents.
  - Tampering independent of fire (learning that one is true would not change your beliefs about the probability of the other)
  - Alarm depends on both Fire and Tampering: it could be caused by either or both
  - Smoke is caused by Fire, and so is independent of Tampering and Alarm given whether there is a Fire
  - Leaving is caused by Alarm, and thus is independent of the other variables given Alarm
  - Report is caused by Leaving, and thus is independent of the other variables given Leaving

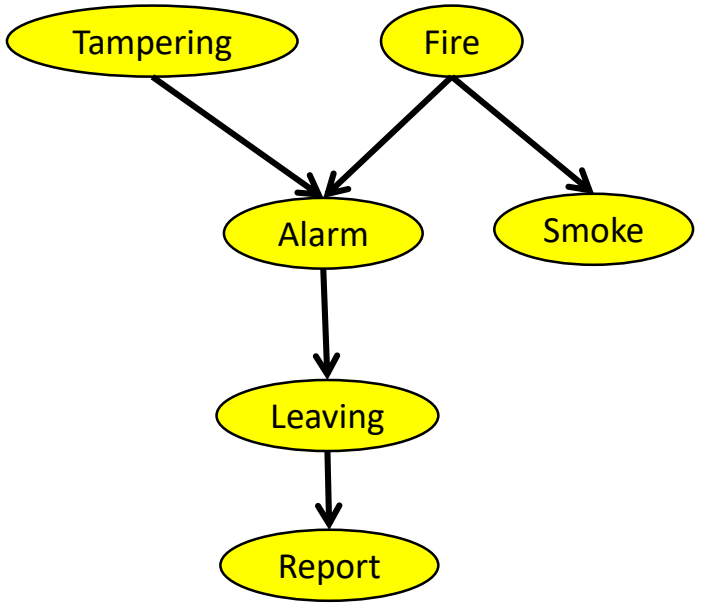


Example for BN construction: Fire Diagnosis



P(Tampering, Fire, Alarm, Smoke, Leaving, Report) =
P(Tampering) x P(Fire) x P(Alarm|Tampering, Fire) x P(Smoke|Fire) x P(Leaving|Alarm) x P(Report|Leaving)

Example for BN construction: Fire Diagnosis

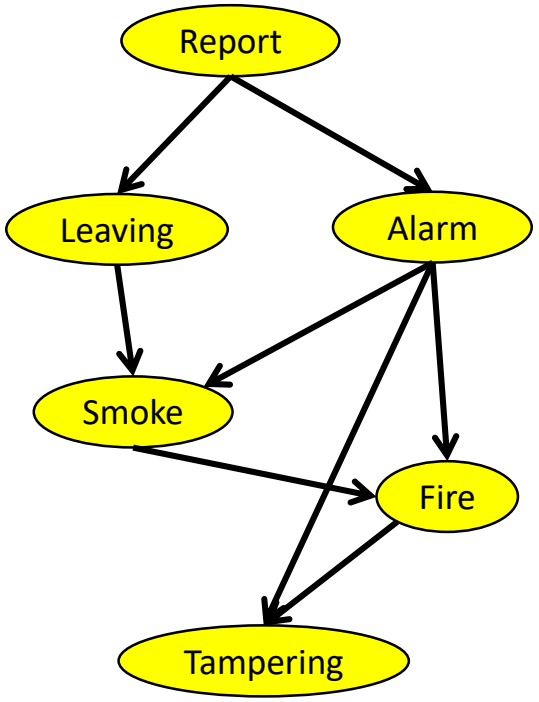


- How many probabilities do we need to specify for this Bayesian network?
  - 1+1+4+2+2+2 = 12

BN construction

- Order matters!
- The complexity of the BN depends on the order in which the variables are introduced
- A “bad” order -> many unnecessary links
  - Not exploiting conditional independence as much

Example for BN construction: Fire Diagnosis



- How many probabilities do we need to specify for this Bayesian network?
  - 1+2+2+4+4+4 = 17

Classwork: Burglar Alarm

- I have a burglar alarm that is sometimes set off by minor earthquakes. My two neighbors, John and Mary, promised to call me at work if they hear the alarm
  - Example inference task: suppose Mary calls and John doesn't call. What is the probability of a burglary?
- What are the random variables?
  - *Burglary, Earthquake, Alarm, JohnCalls, MaryCalls*
- Create a Bayesian Network (only the graph) for this application

Classwork: Burglar Alarm

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  - Example inference task: suppose Mary calls and John doesn't call. What is the probability of a burglary?
- What are the random variables?
  - *Burglary, Earthquake, Alarm, JohnCalls, MaryCalls*
- Create a Bayesian Network (only the graph) for this application
- What are the direct influence relationships?
  - A burglar can set the alarm off
  - An earthquake can set the alarm off
  - The alarm can cause Mary to call
  - The alarm can cause John to call

Example

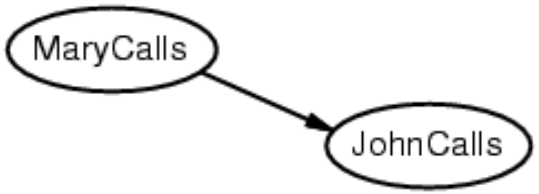
- Suppose we choose the ordering M, J, A, B, E

P(J | M) = P(J)?

Example

- Suppose we choose the ordering M, J, A, B, E

P(J | M) = P(J)? No



Example

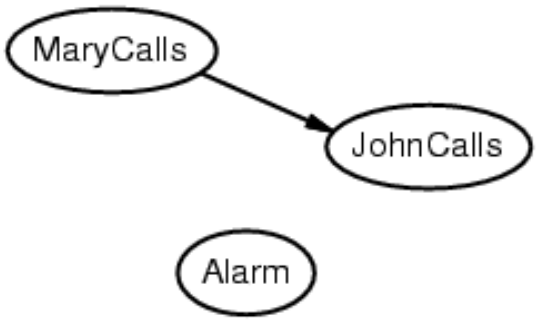
- Suppose we choose the ordering M, J, A, B, E

P(J | M) = P(J)? No

P(A | J, M) = P(A)?

P(A | J, M) = P(A | J)?

P(A | J, M) = P(A | M)?



Example

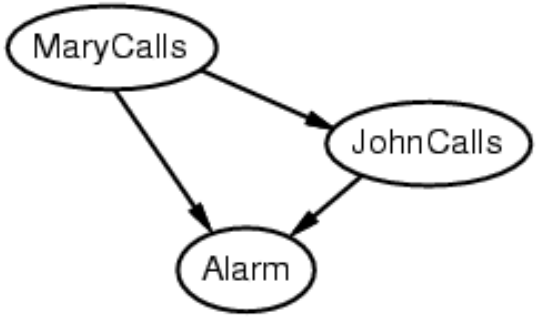
- Suppose we choose the ordering M, J, A, B, E

P(J | M) = P(J)? No

P(A | J, M) = P(A)? No

P(A | J, M) = P(A | J)? No

P(A | J, M) = P(A | M)? No



Example

- Suppose we choose the ordering M, J, A, B, E

P(J | M) = P(J)? No

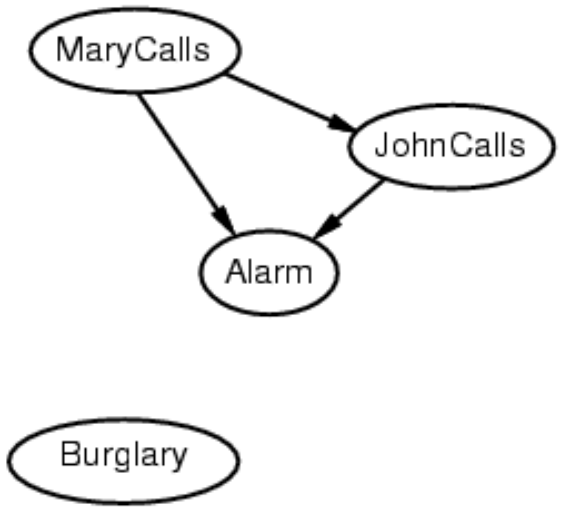
P(A | J, M) = P(A)? No

P(A | J, M) = P(A | J)? No

P(A | J, M) = P(A | M)? No

P(B | A, J, M) = P(B)?

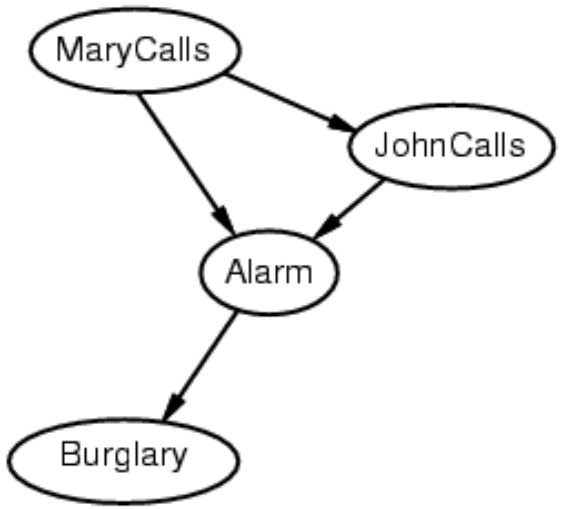
P(B | A, J, M) = P(B | A)?



Example

- Suppose we choose the ordering M, J, A, B, E

P(J | M) = P(J)?		No
P(A | J, M) = P(A)?		No
P(A | J, M) = P(A | J)?	No	
P(A | J, M) = P(A | M)?		No
P(B | A, J, M) = P(B)?	No	
P(B | A, J, M) = P(B | A)?		Yes



Example

- Suppose we choose the ordering M, J, A, B, E

P(J | M) = P(J)? No

P(A | J, M) = P(A)? No

P(A | J, M) = P(A | J)? No

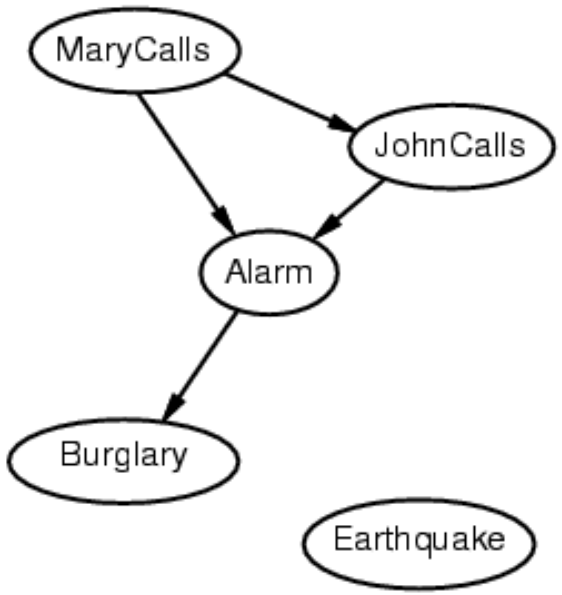
P(A | J, M) = P(A | M)? No

P(B | A, J, M) = P(B)? No

P(B | A, J, M) = P(B | A)? Yes

P(E | B, A ,J, M) = P(E)?

P(E | B, A, J, M) = P(E | A, B)?



Example

- Suppose we choose the ordering M, J, A, B, E

P(J | M) = P(J)? No

P(A | J, M) = P(A)? No

P(A | J, M) = P(A | J)? No

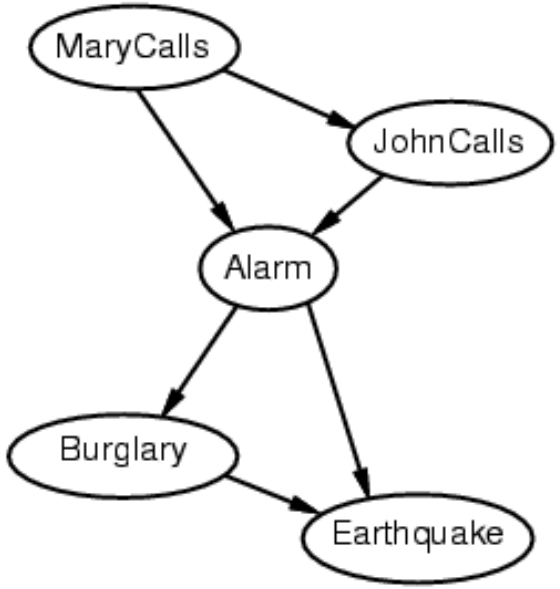
P(A | J, M) = P(A | M)? No

P(B | A, J, M) = P(B)? No

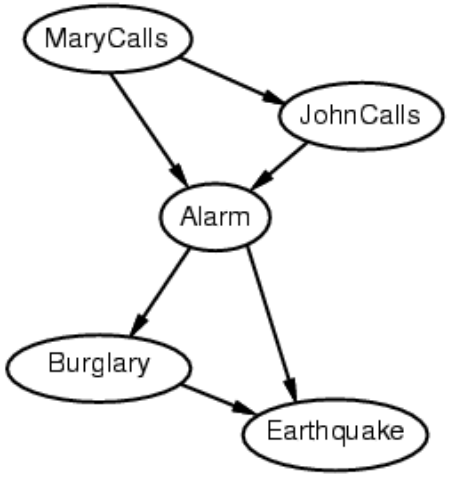
P(B | A, J, M) = P(B | A)? Yes

P(E | B, A ,J, M) = P(E)? No

P(E | B, A, J, M) = P(E | A, B)? Yes



Example: The network we constructed

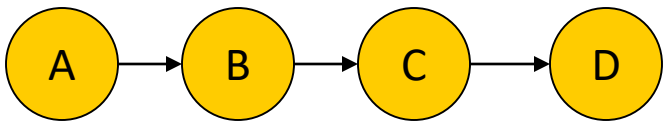


- Deciding conditional independence is hard in non-causal directions
  - The causal direction seems much more natural
- Network is less compact: 1 + 2 + 4 + 2 + 4 = 13 numbers needed

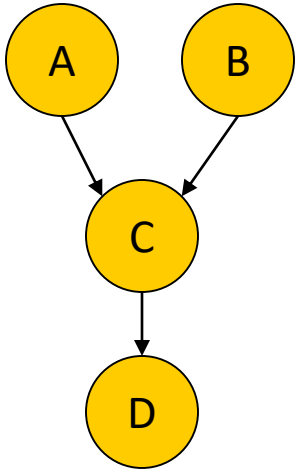
Learning Bayesian Networks from Data

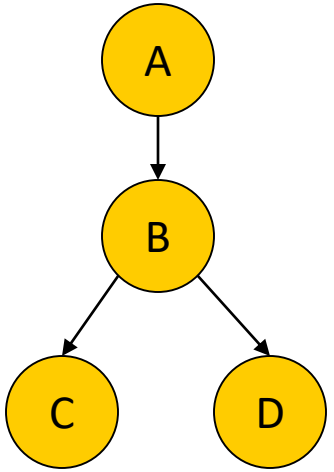
Given a data set, which Bayesian network with variables A, B, C and D best represents the data?

A	B	C	D
true	false	false	true
true	false	true	false
true	false	false	true
false	true	false	false
false	true	false	true
false	true	false	false
false	true	false	false
:	:	:	:



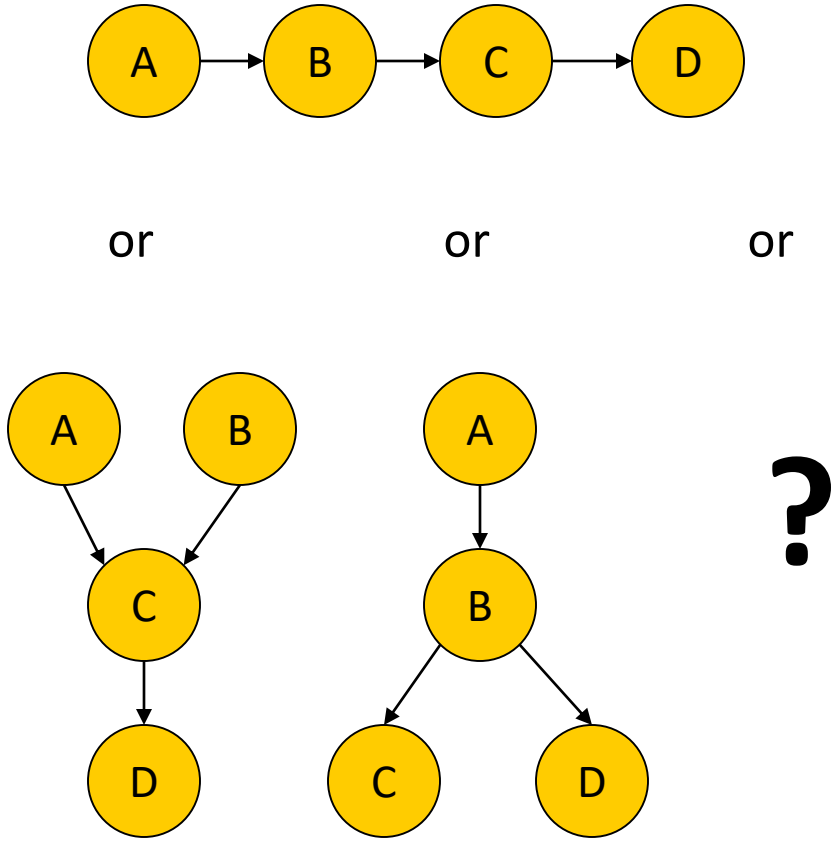
or or or





?

Learning Bayesian Networks from Data



- Each possible structure contains information about the conditional independence relationships between A, B, C and D
- We would like a structure that contains conditional independence relationships that are supported by the data
- Note that we also need to learn the values in the CPTs from data
- *Still an open problem*

Bayesian Networks summary

Two important properties:

1. Is a compact representation of the joint probability distribution over the variables
2. Encodes the conditional independence relationships between the variables in the graph structure

References

- Russel and Norvig, Artificial Intelligence: A Modern Approach, 4th edition, Pearson, 2020.

  Chapter 13.1-13.4